

Short-form video platforms drive mobile usage

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Significance

Short-form video platforms, such as TikTok, have revolutionized digital media, raising concerns about compulsive use and its consequences for well-being. While prior work links both how and what we use our phones for to negative outcomes, it is unclear whether content formats causally shape usage patterns. Using a quasi-experimental design and broadly representative U.S. data, we show that short-form video platforms increase total mobile use, shorten breaks between sessions, and spill over into social media and retail domains, with effects varying across demographic groups but not extending into nighttime hours. These findings indicate that short-form video platforms act as a structural driver of mobile usage, reinforcing habitual checking and highlighting opportunities for interventions that curb harmful engagement.

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Abstract

Short-form video platforms, such as TikTok, have rapidly transformed digital media consumption, with billions of active users worldwide. Their design features, such as algorithmic personalization, endless scrolling, and rapid novelty, have raised concerns about mobile overuse and its associated adverse effects on well-being. However, evidence on whether short-form videos increase mobile usage remains limited. To address this gap, we estimate the causal effect of short-form video platforms on mobile usage behavior using individual-level data ($N = 1,764$) drawn from a representative panel of U.S. mobile device users. Specifically, we compare TikTok adopters to adopters of Instagram/Facebook during the period (1/5/2019–5/8/2020) when short-form video was the key distinguishing feature of TikTok (i.e., before these competitors introduced their own short-form video products), thereby shedding light on the effect of exposure to short-form videos. We find that using short-form video platforms significantly increases total mobile usage duration by up to 17% (about 28 minutes per day for the average user) and reduces average time away from the mobile phone by -20% (29 fewer minutes per day for the average user), but does not consistently affect the number of daily sessions. Effects are disproportionately higher among individuals who previously used their phones less, relative to those with higher baseline usage. Contrary to fears about nocturnal overuse, effects are only significant during daytime hours. Surprisingly, older individuals have significantly larger increases in mobile sessions per day from usage of short-form video platforms compared to younger individuals. Altogether, our results show a behavioral mechanism through which short-form video platforms contribute to mobile usage, offering new insights into theories of digital addiction and highlighting the importance of promoting healthier digital engagement.

Keywords: short-form video platforms, social media, addiction, mobile usage

1 Main

Short-form videos have reshaped the social media landscape and now play a substantial role in how individuals engage with digital media [1]. TikTok, which first popularized the format, has more than 1.59 billion monthly active users, about one in five people worldwide [2]. In a few years, it has grown into one of the largest global social media platforms, particularly popular among adolescents [3]. The rapid rise of TikTok has also shaped industry-wide strategies, prompting competitors such as Instagram, Facebook, and YouTube to launch and heavily prioritize their own short-form video products.

There are growing concerns that the widespread exposure to short-form video platforms encourages compulsive engagement and overuse [1]. Platform features aimed at maximizing engagement, such as algorithmic personalization, endless scrolling, and rapid novelty, create conditions that may facilitate addictive behavior ‘by design’ [4,5,6,7,8,9]. Building on these concerns about platform features, a growing body of research links frequent consumption of short-form videos to a range of adverse outcomes. For example, such users often experience distorted time perception [9,10], reduced sustained attention [11,12], essential hypertension [13], and impaired prospective memory performance due to rapid context switching [14]. Beyond these cognitive effects, short-form videos can also shape behavior in other domains, for example, by stimulating impulse buying and purchase intentions [15,16]. Furthermore, excessive consumption of short-form videos may negatively shape mental well-being, including life satisfaction [17,18,19] and depression [20].

These concerns extend previous work indicating that mobile overuse, more generally, is consistently associated with adverse outcomes across mental and physical well-being [21,22]. Mobile overuse predicts mental health challenges such as lower life satisfaction and increased psychological distress, especially among adolescents [23,24,25,26,27,28,29]. Further, excessive social media use can impair daily functioning and work performance [30]. Among adolescents, nighttime mo-

bile use is linked to sleep disturbances [31], which, in turn, contribute to depressive symptoms [32].

The above works indicate overall negative effects of mobile overuse and excessive consumption of short-form videos [9, 12, 21, 22], but the mechanisms through which short-form video platforms contribute to mobile overuse remain unclear. One possibility is that short-form video platforms primarily heighten risks indirectly, by extending time spent on mobile devices and thereby reinforcing patterns of overuse. Another is that the platforms introduce distinct risks through their unique design features that impair attention and well-being even without increasing overall use. A third is that short-form video platforms reshape the timing and rhythm of mobile activity, for example, by promoting nighttime use that disrupts sleep. Disentangling the underlying mechanism is important, as it is otherwise difficult to determine whether short-form video platforms merely reflect a broader trend in mobile content or represent a structural driver of mobile usage. Yet, there is little causal evidence that disentangles these possibilities in the case of platforms like TikTok.

To address these gaps, we estimated the causal effect of short-form video platforms on mobile usage behavior. Our analysis was based on individual users ($N = 1,764$) drawn from a representative panel of U.S. mobile device users, which allowed us to track the complete history of app and website activity at the individual level. We focused on three key metrics of mobile usage: (1) duration, (2) number of sessions, and (3) average time away from the mobile phone (TAP), which together capture the intensity, frequency, and (dis)engagement of mobile usage. We used a quasi-experimental difference-in-differences (DiD) design with matching to estimate the effect of short-form video platforms. Hence, our identification relied on exploiting the parallel trends assumption, i.e., that, in the absence of short-form video platforms, the treatment and control groups would have evolved similarly over time. Crucially, this approach allows for time-invariant unobserved confounders, as their contributions are canceled out through the differencing based on the parallel trends assumption. Treatment and control groups were constructed to be comparable in the size and characteristics of the platforms they adopted, differing only in exposure to short-form video. Specifically, individuals who adopted TikTok served as the treatment group, while

those who adopted Instagram or Facebook during a period before the introduction of their own short-form video formats served as the control group. Because short-form video was the primary distinguishing feature of TikTok relative to Instagram and Facebook at that time, this design isolates the effect of exposure to short-form video on mobile usage behavior.

We found that short-form video platforms led to sustained increases in overall mobile usage. Total mobile duration increased by 8% within the first week of exposure and reached a 17% increase after nine weeks, which corresponds to 28 minutes more mobile usage per day for individuals adopting short-form video platforms. Similarly, the average TAP decreased by 20%, corresponding to an average of 29 minutes less between sessions. Short-form video platforms had the largest effect on individuals with initially lower mobile usage, reaching more than 30% increased mobile usage duration. We found that the effects are concentrated during daytime hours, but with no evidence of increased mobile use at night. While daily session counts showed no significant increases on average, we observed substantial variation across sociodemographic subgroups, with older individuals exhibiting notably larger increases in daily mobile sessions. Finally, these increases in usage can be attributed to higher engagement with social media domains in general, where duration rose by 45%. The consumption of retail domains has also increased significantly due to short-form video platform use, with a total duration increase of 30%. As a sanity check, we repeated our analysis for a later period when Instagram and Facebook had already introduced their own short-form video formats. In this setting, the differences between TikTok adopters and control users vanished, indicating that the observed effects stem from the short-form video format rather than other platform-specific features.

Altogether, our findings bridge concerns about short-form video platforms and the harms of increased mobile usage by showing evidence that the former causally drove the latter. This suggests that the risks of short-form video platforms, therefore, extend beyond content effects (what individuals consume) to structural effects on mobile usage (how much individuals use their mobiles). Specifically, we showed that short-form video platforms reinforced mobile usage by increasing

usage durations and shortening the intervals between sessions. This has direct implications for platform designers, policymakers, and mental health professionals, who face the challenge of mitigating these risks. While first efforts to curb excessive digital engagement are emerging, our results highlight the need for more comprehensive strategies that address the unique design features of short-form video platforms and their potential to intensify use. For example, strategies that introduce friction into use, such as reminders to take breaks, time caps on feeds, or nudges to disengage, have already been shown to reduce endless scrolling and improve mood in other settings [33,34]. Tailoring such approaches to the unique format of short-form video could help limit harmful overuse while preserving the benefits of digital connectivity.

2 Results

2.1 Overview

Our analysis draws on a sample ($N = 1,764$) drawn from a large representative ($>20,000$) of U.S. mobile device users during the period when TikTok was the only platform offering short-form video (1/5/2019–5/8/2020). Collected by the external media analytics company Comscore, our panel data provides a complete history of all web domains visited by individuals on their mobile devices, including both apps and websites. To estimate the effect of short-form video platforms, we compared individuals who adopted TikTok (i.e., 1,160 individuals) to individuals who adopted Instagram or Facebook (i.e., 604 individuals) before the latter introduced their own format of short-form videos.

For each individual, we collected their full mobile usage history spanning 60 days before and after the adoption window, allowing us to capture changes in mobile usage post-adoption. We defined adoption as the first visit to a platform’s app followed by at least one additional visit within a 10-day window, thereby excluding cases where individuals installed an app but did not continue

using it. For external validity, we compared the demographic distribution of our filtered sample to that of the full Comscore panel, which aims to be representative of the US over-18 population (Supplementary Figure S2). Our selected sample is broadly comparable to the overall Comscore panel. Results are robust to alternative adoption definitions; see Supplementary Tables S8 to S10.

To estimate the effect of short-form video platforms on mobile usage metrics, we used TikTok adopters as our treatment group and compared them to Instagram or Facebook adopters as our control group before the latter introduced their own format of short-form videos (see Figure 1a). To understand how adoption reshaped usage patterns over time, we focused on three main metrics: (1) duration, (2) number of sessions, and (3) average time away from the mobile phone (TAP). To ensure that treatment and control groups were comparable in their phone usage before the intervention, we applied propensity score weighting based on pre-adoption mobile usage across the three metrics above [35,36] (see Figure 1b).

We then estimated the causal effect of adopting a short-form video platform (TikTok) rather than traditional social media (Instagram/Facebook) using a stacked, dynamic difference-in-differences (DiD) design [37, 38] with propensity score weighting (see Figure 1c). This methodological choice is motivated by three considerations. First, this DiD design leverages the staggered adoption by stacking individuals into subexperiments based on their adoption week. Second, it allows us to estimate dynamic effects for different time periods before and after adoption, which captures how effects evolve over time. Importantly, the dynamic effect estimates allowed us to assess whether treated and control groups followed parallel trends before adoption, providing a test of the parallel trends assumption, a key identification assumption for the validity of DiD [39]. Importantly, the parallel trends assumption eliminates the strong assumption of “no unobserved confounders” within observational studies, as long as those confounders are time-invariant. Third, propensity score weighting addresses concerns regarding how individuals from the treatment and control groups differ in terms of their pre-treatment mobile usage. More details on the empirical model are provided in Section 4.

2.2 Overall effect of short-form video platforms

Short-form video platforms led to significant changes in mobile usage behavior (see Figure 1c). Notably, the total duration on mobile increased substantially due to short-form video platforms. Already after one week, mobile usage duration was 7.52% higher relative to the expected duration in the absence of the short-form video format. This effect grew steadily over time, reaching a 17.44% increase after nine weeks ($p < 0.001$). In absolute terms, this corresponds to roughly 28 additional minutes of mobile use per day, compared to the pre-adoption mean of 2:43 hours for individuals in the treatment group. To put this in perspective, the effect of adopting short-form video platforms on usage duration is almost as large as the pre-adoption gender gap in mobile usage duration and amounts to roughly one-third of the difference between young and older individuals (see Supplementary Table S7). In contrast, the effect on the number of mobile sessions was not consistently statistically significant. Finally, the average TAP showed a clear decrease. After nine weeks, the average TAP of individuals using short-form video platforms reduced by 20% ($p < 0.001$). This drop corresponds to approximately 29 minutes less time between sessions, compared to the pre-adoption average TAP of 2:25 hours for individuals in the treatment group. Again, this decline is comparable in magnitude to the pre-adoption gender difference in avg TAP and less than half the age difference (see Supplementary Table S7). These findings provide strong evidence that consumption of short-form video platforms causally changes mobile usage patterns by increasing total duration and reducing average TAP, while effects on the number of sessions remain non-significant.

We examined the robustness of our findings in the following ways. First, we explored the plausibility of the parallel trends assumption using a pre-test on the pre-adoption effect estimates (see Supplement C.1 for more details on this test). We found no significant differences between treatment and control groups before adoption for all three metrics (see Table S1), supporting the validity of our selected DiD design. Second, we compared TikTok adopters to a set of different control groups consisting of (i) non-adopters of TikTok and (ii) adopters of Instagram and Face-

book during a period *after* the latter platforms introduced short-form videos (see Supplement C.2). Importantly, we found no effect in the latter scenario, meaning that once the short-video format was widespread, the difference in subsequent outcomes between adopting TikTok rather than Instagram/Facebook vanished. This result provides strong evidence that the estimated causal effects in our identification strategy were primarily driven by the exposure to short-form videos, rather than other platform-specific features or the intrinsic qualities of TikTok (e.g., friendship networks or particularly effective algorithmic personalization). Third, we evaluated the robustness of our findings using alternative model specifications that are commonly used in the literature (e.g., [40, 41, 42]), namely, the standard two-way fixed effects DiD design and the interaction-weighted estimator [43] (see Supplement C.3). The results show that the effect estimates are not sensitive to the choice of the model specification and corroborate our findings from above. Fourth, we performed a causal sensitivity analysis to examine the potential influence of time-varying unobserved confounders (see Supplement C.4.1). The results indicate that any such confounder would need to exhibit a joint association of approximately 11.08% with both adoption choice and mobile usage duration (and 13.98% with average TAP) to meaningfully alter the estimated effects — a magnitude we judge to be implausible in this context, given that even age explains only 3.31% (and 2.12%) of the respective variances. Fifth, we performed a placebo test in which we artificially shifted the adoption week to several weeks before the actual adoption to assess potential anticipation effects (see Supplement C.4.2). The absence of systematic effects in these placebo periods suggests that the observed post-adoption changes were not driven by pre-adoption behavioral adjustments.

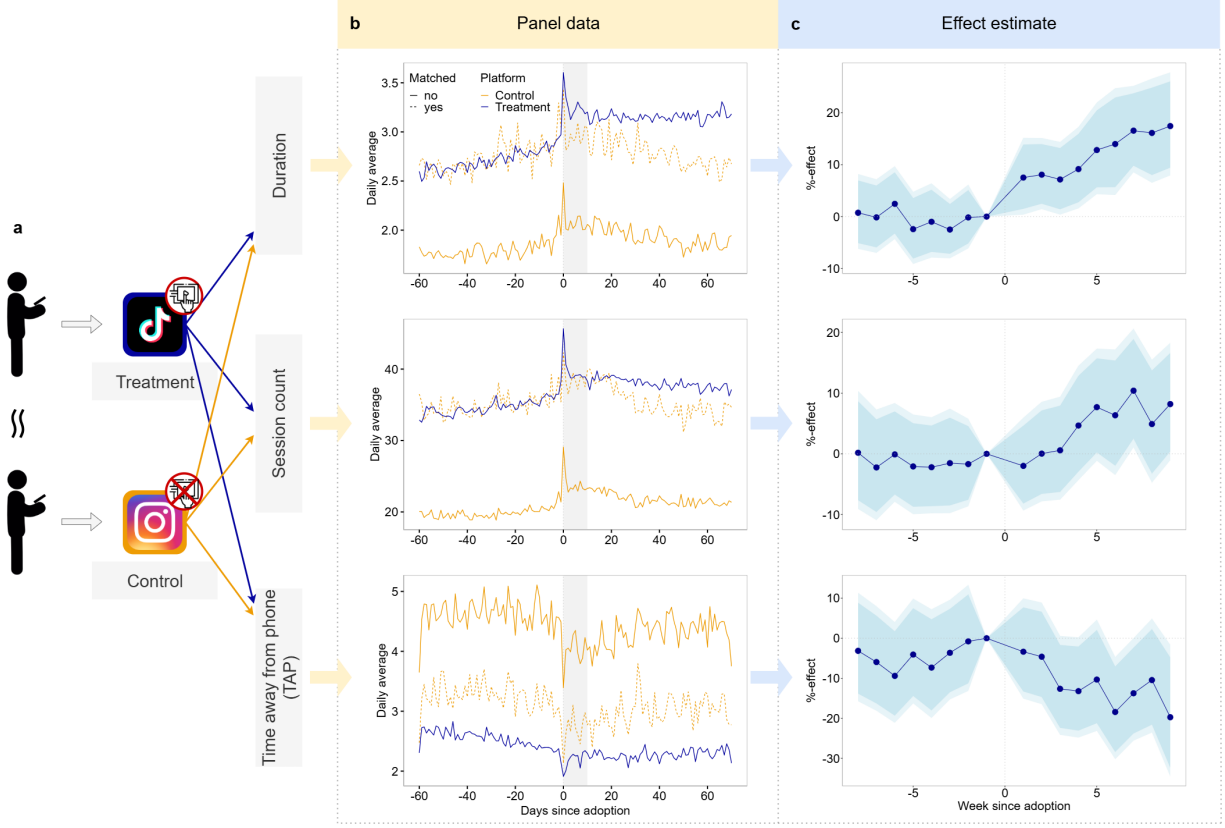


Figure 1: Estimated effect of short-form video platforms on mobile use. **a**, We estimated the effect of short-form video platforms by carefully constructing comparable groups of TikTok adopters (treatment group) and Instagram or Facebook adopters before the two platforms introduced their format of short-form videos (control group). Hence, short-form videos were the main distinguishing feature between the treatment and control groups, and the observed differences in mobile usage behavior can therefore most plausibly be attributed primarily to exposure to short-form videos. We constructed panels on the resulting $N = 1,764$ individuals (with 1,160 in the treatment group and 604 in our control group) based on three different mobile usage metrics: (1) duration, (2) session count, and (3) average time away from mobile phone (TAP). **b**, We applied propensity score weighting [35, 36] based on pre-adoption mobile usage to ensure that treatment and control groups are comparable. The figure illustrates the three outcomes of interest for both matched and unmatched panels. **c**, The plots show the estimated effects of exposure to short-form videos on mobile use. We estimated the effects using a stacked, dynamic difference-in-differences (DiD) design [37, 38]. Importantly, by using a leads-and-lags specification, the estimated dynamic treatment effects (shown by dark blue points) enabled us to perform a pre-test of the parallel trends assumption and to assess how the effects evolved post-adoption. The light (dark) blue shaded areas indicate the 90% (95%) confidence intervals. We report the effect estimates for each metric in Supplementary Table S1.

2.3 Low- vs. high-intensity mobile users

We analyzed whether the effect of short-form video platforms differs between individuals with low and high baseline mobile usage intensity. To construct these groups, we computed the median daily mobile usage duration during the pre-adoption period and classified low-intensity or high-intensity users based on whether they fell below or above the sample median. We then estimated effects separately for both groups using the stacked, dynamic DiD framework. This allowed us to capture potential differences in how short-form video platforms affect individuals who were already high-intensity mobile users compared to those with relatively lower baseline usage.

Short-form video platforms increased mobile usage for both low- and high-intensity users, but the magnitude of effects differed markedly across groups (see Figure 3). For low-intensity users, the effect on mobile usage duration was 19.20% ($p < 0.001$) already one week post-adoption and increased to 31.43% ($p < 0.001$) after nine weeks, while, for high-intensity users, the increase was 14.00% ($p < 0.001$) after nine weeks. Again, the effect on the number of mobile sessions was not consistently statistically significant for both groups. However, the average TAP showed a clear decrease for low-intensity users (nine weeks post-adoption: -22.44% ; $p < 0.001$), while high-intensity users did not significantly reduce their average TAP.

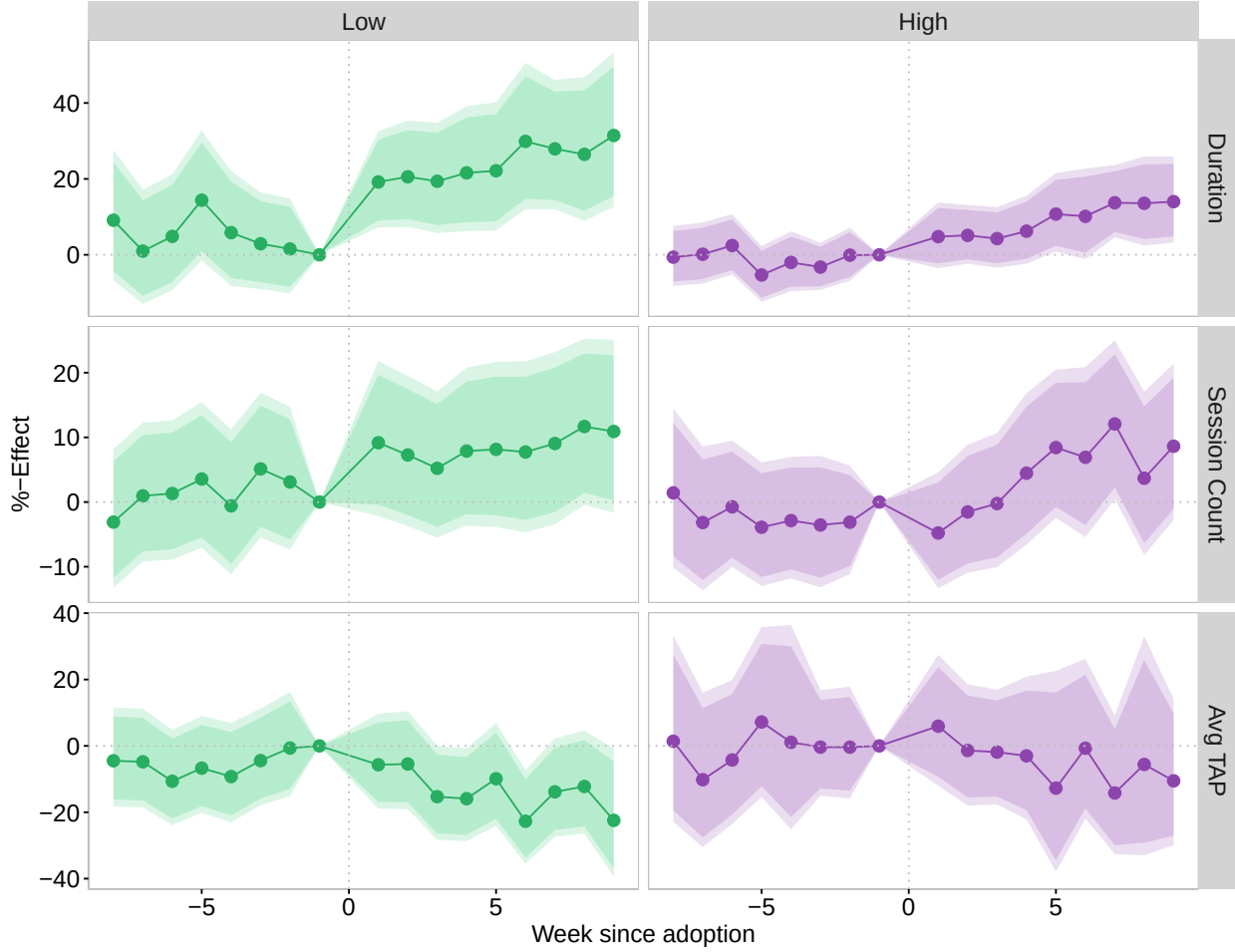


Figure 2: **Effect estimates (in %) of short-form video platforms for low vs. high intensity mobile users.** Estimates are based on the stacked, dynamic DiD framework applied separately to low vs. high-intensity mobile users. We estimated the median daily duration (pre-adoption) of all individuals in our panel and then classified them into equally sized groups based on whether they were above or below this median. The light (dark) shaded areas indicate the 90% (95%) confidence intervals. We report the effect estimates and sample sizes for low vs. high-intensity users in Supplementary Table S2.

2.4 Daytime vs. nighttime effects

We separated the panels for daytime and nighttime mobile usage to examine whether short-form video platforms affected activity differently across these periods. Prior research showed that social media use is often linked to delayed sleep, reduced sleep quality, and adverse health consequences [31]. Therefore, understanding whether short-form video platforms increase nighttime activity helps determine if such platforms contribute to sleep displacement and related harms. We followed prior research [44] and used our metrics as proxies for sleep quality during nighttime. To capture individual differences in sleeping patterns, we constructed a fixed 10-hour nighttime window for each individual around their median longest away time from mobile recorded for each day. This window is designed to capture not only the presumed sleep period but also the minutes before falling asleep and immediately after waking up, ensuring a comprehensive measure of nighttime use. The remaining hours are classified as daytime.

We find no evidence that usage of short-form video platforms led to an increase in mobile usage during nighttime hours compared to the usage of social media in general (see Figure 3). Specifically, our estimates show that the effects were concentrated during the day, with no significant changes in either duration or session frequency observed at night. Our findings suggest that, while social media in general has been linked to disrupted sleep patterns and nocturnal overuse, short-form video platforms appear to have primarily increased mobile usage during unstructured daytime hours, rather than additionally disrupting sleep at night.

We conducted robustness checks using narrower windows of 8 and 6 hours to ensure that our results are not driven by the definition of the nighttime window. The results remain consistent (see Supplementary Tables S4 and S5), with no significant effects during nighttime for the three mobile usage metrics in either specification.

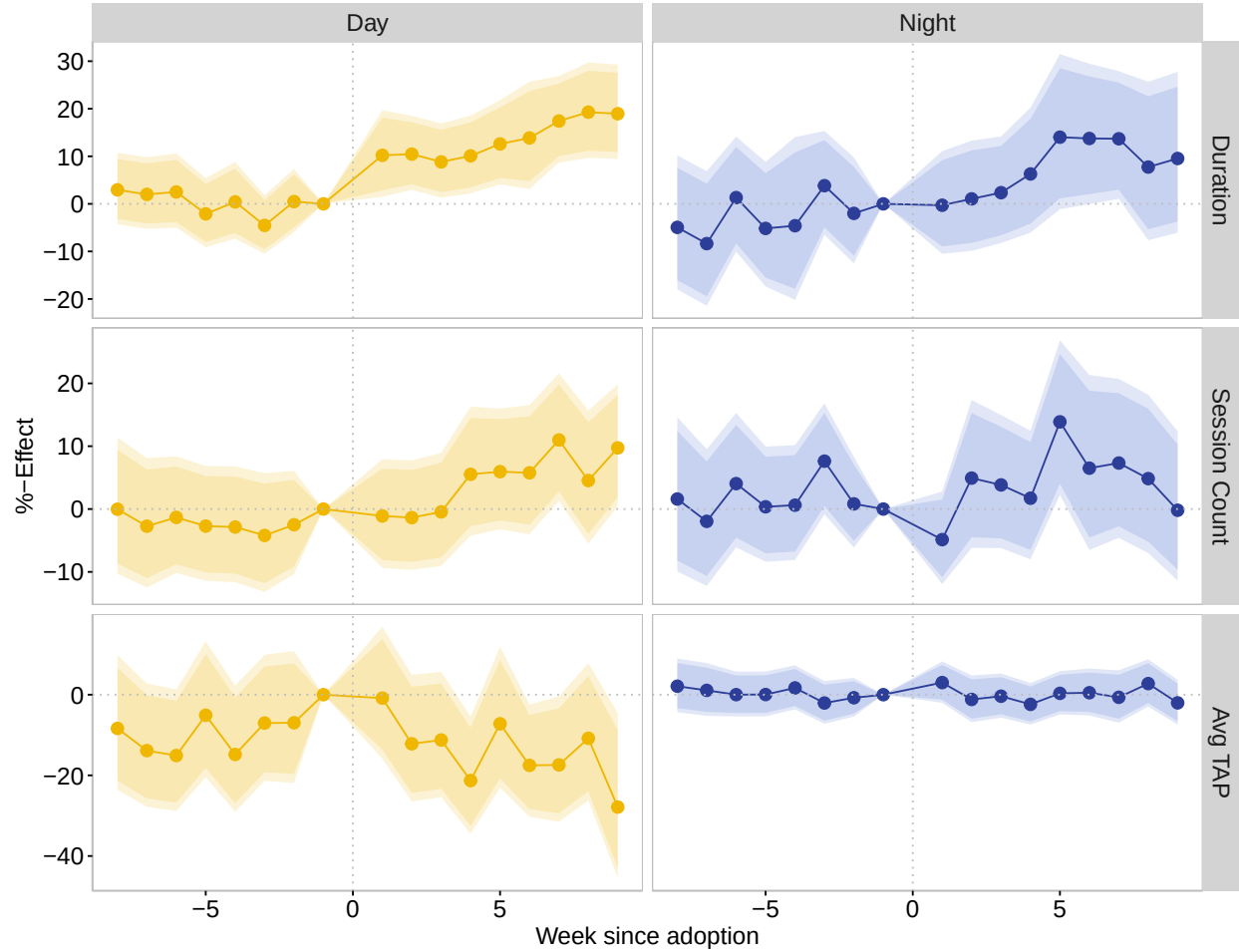


Figure 3: Effect estimates (in %) of short-form video platforms during the day vs. night. Estimates are based on the stacked, dynamic DiD design applied separately to day vs. nighttime. For each individual, nighttime is defined as a fixed 10-hour window around their median longest away time from their mobile across the days in the panel, with all remaining hours classified as daytime (see Section 4). This individualized definition ensures that the nighttime window reflects the typical sleep hours of each individual. The light (dark) shaded areas indicate the 90% (95%) confidence intervals. We report the effect estimates and sample sizes for each day vs. nighttime in Supplementary Table S3.

2.5 Effect heterogeneity by sociodemographics of individuals

Overall, we found substantial differences in the effects of short-form video platforms across most sociodemographic subgroups (Figure 4), which reveals who is most at risk of increased mobile usage due to short-form video platforms. While females spent more time on their phones due to short-form video platforms than males, males had significantly more mobile sessions per day ($p < 0.05$). Notably, the effect on the number of sessions varied significantly across most sociodemographic subgroups. Hence, while the average effect across the entire population showed no consistent increase in session count, certain subgroups were more impacted and significantly increased their session counts post-adoption, while others did not. For example, older individuals had a significantly higher effect on session count than middle-aged ($p < 0.01$) and young individuals ($p < 0.05$). In absolute terms, this translates to 2.89 additional daily sessions for older individuals relative to their pre-adoption baseline of 30.14 sessions per day, whereas, for middle-aged and younger individuals, the number of sessions per day decreased by 2.45 and 0.16, respectively, from their baselines (i.e., 40.2 and 36.9, respectively). Finally, for average TAP, we also observe variation across subgroups, although the patterns are less uniform compared to those for duration and session count. These results show that the effect of short-form video platforms was not uniform across sociodemographic subgroups.

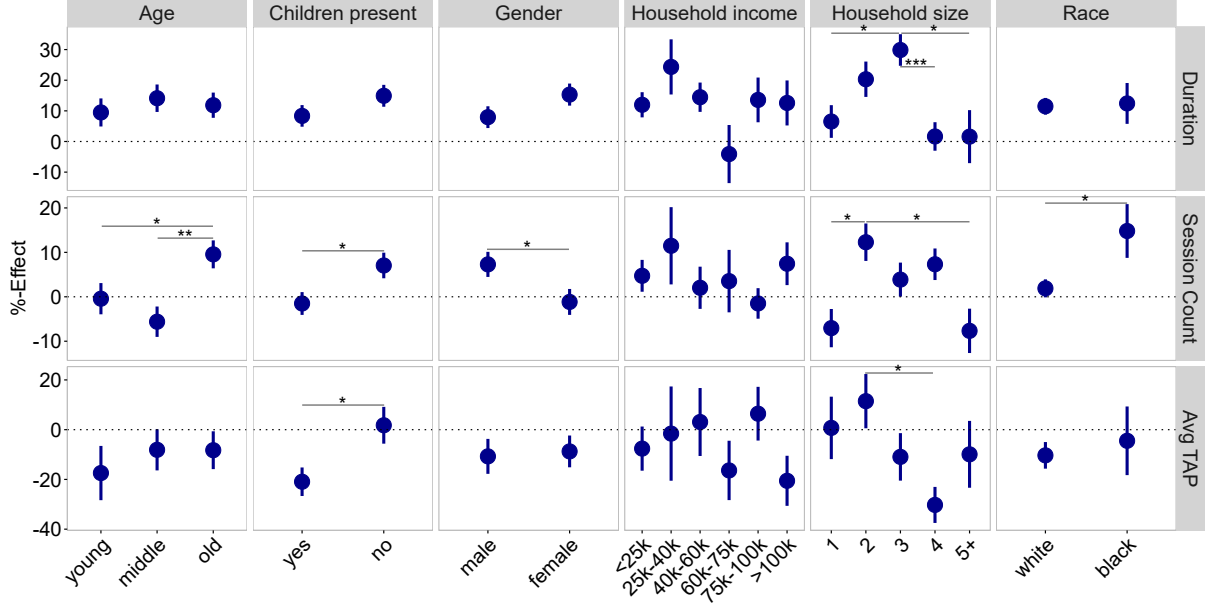


Figure 4: **Post-adoption effect estimates (in %) of short-form video platforms across different sociodemographic subgroups.** Estimates are based on the stacked, dynamic DiD framework applied separately to each subgroup. We aggregated post-adoption effects using an averaging as proposed in [43] (see Section 4 for more details) together with bootstrapped standard errors ($b = 2,000$) to quantify uncertainty and to test for statistically significant differences between subgroups using a t -test. To account for multiple comparisons, p -values from pairwise subgroup tests were adjusted using the Holm–Bonferroni procedure [45]. We report the effect estimates, standard errors, and number of individuals for each sociodemographic subgroup in Supplementary Table S6.

3 Discussion

The rise of short-form video platforms such as TikTok has raised concerns about the downstream adverse effects of short-form video overuse on mental health outcomes (e.g., [17, 18, 19]). More generally, mobile overuse itself has been linked to lower life satisfaction, greater psychological distress, impaired daily functioning, and sleep disturbances that contribute to depressive symptoms [21, 22, 23, 31, 32]. Yet, the relationship between harms associated with mobile usage behavior (e.g., late at night, for extended periods) and harms associated with specific types of content (e.g., short-form videos) remains unclear. We addressed this gap by shedding light on the effect of exposure to short-form videos on broader mobile usage behavior. Based on a complete history of app and website activity $N = 1,764$ drawn from a large sample of U.S. mobile device users, we estimated the effect of short-form video platforms on mobile usage behavior using a quasi-experimental design (DiD). Our key finding is that short-form video platforms increased users’ total mobile usage by 17% on average and decreased the average time away from the mobile phone (TAP) by 19% on average. These effect sizes continued to increase with time, in a trajectory consistent with habit formation theory and behavioral addiction [4, 46, 47].

A central feature of our empirical strategy is the comparison of TikTok adopters to those of Instagram and Facebook during a period when TikTok was the only large social media platform offering short-form videos. Although other platform aspects (e.g., differences in friendship networks or algorithmic personalization) could contribute to the effects, our evidence suggests that short-form video is the primary driver. First, this was the key distinguishing feature at the time of the study, when Instagram and Facebook had not yet introduced their own short-form video products. Second, many users interact with overlapping friendship networks and influencers across platforms, and the recommendation algorithms of major social media platforms are generally built around similar reward systems designed to maximize engagement, making it unlikely that these factors alone explain the observed patterns. Third, our additional analyses reinforce this interpre-

tation: when we re-estimated the effects using control groups from a later period in which Instagram and Facebook had launched short-form video, we found no significant differences between TikTok adopters and control group users (see Supplement C.2). The disappearance of effects once short-form videos became widely available strongly supports the interpretation that our estimates primarily capture the effect of exposure to short-form videos rather than platform-specific characteristics or other intrinsic qualities of TikTok. Taken together, these findings support the view that it is the introduction of short-form video itself, not other platform features, that drives the observed increase in mobile usage.

Central to our identification strategy was the exploitation of the adoption event under the parallel trends assumption. This assumption implies that, in the absence of treatment, the mobile usage behavior of TikTok adopters and the control group (Instagram/Facebook adopters) would have followed similar trends over time. Here, unlike many observational studies that rely on a strict “no unobserved confounders” assumption, our DiD framework allows for unobserved confounding as long as it is time-invariant. Specifically, any variable, such as personality traits or baseline preferences for digital media use, that might influence both platform adoption and mobile engagement were differenced out through the exploitation of the parallel trends assumption. We followed best practices by testing the plausibility of the parallel trends assumption through pre-tests, which showed no significant coefficient estimates in the pre-adoption periods and thereby support the validity of our identification strategy (see Supplement C.1). Moreover, the presence of time-varying unobserved confounders was unlikely in our setting, as any such confounder would have to show an implausibly strong joint effect on both adoption choice and mobile usage metrics to meaningfully alter the estimated effects (see Supplement C.4.1). In addition, the absence of systematic effects in our placebo tests indicated that the observed post-adoption effects were not driven by anticipation or pre-adoption behavioral adjustments (see Supplement C.4.2).

Prior research links mobile overuse and addiction to disrupted sleep quality, which in turn leads to adverse mental health outcomes [31, 32]. We found no evidence of increased mobile usage at

night from the adoption of a short-form video platform. This could be because nighttime usage patterns are already saturated [32] and therefore show no difference to individuals who adopt Instagram or Facebook. Alternatively, it could be because nighttime use faces natural barriers (e.g., fatigue, sleep routines) [44] that may inhibit additional endless scrolling compared to general content consumption. Individuals may be more prone to low-effort, endless scrolling during the day, when more unstructured moments for mobile usage occur [48,49]. This suggests that interventions aimed at improving sleep quality should focus on reducing *how much* mobile time users engage in rather than targeting *what* users are consuming.

Short-form videos require little effort to consume continuously and foster a hedonic mindset that prioritizes immediate gratification over self-regulation [50], which, in turn, introduces “time-flies” moments and a “just one more video” logic that extends usage beyond intention [7, 51, 52, 53, 54]. Hence, introducing friction into this process could help prevent overuse and thereby minimize risks to well-being. Examples include simple design features such as prompts that ask users to take a break or introduce a short pause before the next video plays. Evidence from related strategies suggests this approach can be effective. For instance, behavioral interventions introduce minimal interruptions before opening applications to disrupt habitual checking, which have been shown to reduce mobile usage [34]. Similarly, randomized interventions that encourage breaks or limit automatic engagement have shown positive effects on well-being [33, 55]. Beyond platform-level designs, boosting the digital literacy of users may provide an additional layer of protection by improving their self-regulation skills. Here, prior work on social media addiction shows that digital literacy can reduce problematic use by helping individuals recognize persuasive design features, set personal limits, and engage with platforms more intentionally [56, 57]. Tailoring such strategies to short-form video platforms may represent a promising pathway to mitigate the risks associated with increased mobile use.

Although our contribution constitutes a meaningful advance by studying the causal effects of consumption on short-form video platforms, it nonetheless has limitations. First, our analysis fo-

cused on a set of individuals in the US, which may limit the generalizability of our findings to other cultural contexts. Second, our outcomes captured mobile usage behavior but do not directly measure addiction, mental health, or physical well-being. However, given well-established evidence linking both mobile overuse and short-form video addiction to adverse outcomes, our findings on usage patterns can be viewed as indicators of potential risks to well-being. Third, our study built on data from a specific timeframe in which Instagram and Facebook had not yet introduced short-form videos. While platform design features may have evolved since then, we argue that the effects of exposure to short-form videos on mobile usage behavior remain consistent, as the underlying engagement mechanisms have not changed. Fourth, our data focused exclusively on mobile phone usage and do not capture consumption on other devices, such as tablets, laptops, or smart TVs. As a result, our estimates may understate total exposure to short-form videos and overall digital media consumption. Fifth, while the Comscore panel is designed to be broadly representative of the U.S. population, full transparency on participation and attrition processes is not available. As such, residual concerns about self-selection into the panel and its comparability to the general population cannot be fully ruled out.

Nonetheless, our results suggest a precise mechanism by which short-form video platforms can increase the risks associated with mobile overuse and, consequently, risks to well-being. As these platforms expand across age groups and become embedded in everyday routines, the potential harms to attention, consumer behavior, and mental health may grow. This highlights the importance for researchers, policymakers, platform designers, and mental health professionals to develop safeguards that curb harmful overuse while preserving the benefits of digital connectivity.

4 Materials and methods

4.1 Data

Mobile usage data. Our analysis is based on data from Comscore, a media analytics company, which passively collects individual-level browsing and application activity data. We used Comscore’s iOS mobile data panel ($N = \sim 21,300$), which has been used in several previous studies [58,59,60,61]. Participants gave informed consent to share their mobile usage data in exchange for monetary compensation and can opt in and out at any point. The dataset contains detailed logs of all apps and websites visited on mobile devices, including precise timestamps and the duration that each individual spent on a given website or app. Additionally, the dataset includes sociodemographic characteristics of individuals who consented to share this information, which includes age, gender, race, parental status, household size, and household income. Each website and app that was accessed was further categorized by Comscore’s Client Focused Dictionary (CFD) into categories and subcategories.¹ For the purposes of this study, we focused on the following categories: social media, retail, entertainment, news/information, services, and games.

Construction of the panel. We constructed our panel to capture a period in which TikTok was the only large social media application offering short-form videos, namely, 1/5/2019 to 5/8/2020.² During this timeframe, the treatment group (i.e., adopters of TikTok) and control group (i.e., adopters of Instagram or Facebook) differed in their exposure to short-form videos, allowing us to shed light on its effect on mobile usage behavior. We applied the following filtering criteria. First, we restricted our panel to individuals who visited the respective social media app for the first time and who (i) then used it at least twice in a 10-day adoption window (to filter out individuals who only installed the application and never used it thereafter), and (ii) had recorded mobile

¹<https://direct-support.comscore.com/hc/en-us/articles/360002188953-Official-Comscore-Media-Matrix-Reporting-Rules>

²The data provided from the media analytics company starts on May 1, 2019, and Instagram introduced its format of short-form videos on August 5, 2020.

activity for at least 60 days prior to that (to filter out individuals who newly entered the panel or had a new mobile device). Second, we excluded individuals who had no mobile activity during the first or last five days of their respective panel to handle panel attrition. Third, we excluded individuals with fewer than 20% active days, as their usage patterns do not provide sufficient information for reliable analysis. Our final sample consisted of $N = 1,764$ individuals, with 1,160 individuals for our treatment group and 604 for our control group. For each individual, we queried their complete mobile activity for the 60 days before and after the 10-day adoption window. We applied several robustness checks in which we varied the above panel characteristics to confirm the robustness of our analyses (see Supplementary Tables S8 to S12). We additionally compared the demographic distribution of our filtered sample to that of the full, representative Comscore panel in Supplementary Figure S2 to show that our selected sample remains broadly comparable to the overall population.

Definition of mobile usage metrics. We defined three main metrics to construct our panel. (1) Total duration of mobile use per day captures the overall engagement length with the device. Duration is a key metric in studies of digital well-being, as higher screen time has been associated with lower life satisfaction and adverse mental health outcomes [25, 62, 63]. (2) The total number of mobile sessions per day measures how often individuals return to their phones. Here, more mobile sessions reflect patterns of frequent checking behavior and were documented as compulsive mobile use in prior research on digital addiction and attention fragmentation [64, 65]. (3) The average time away from mobile phone (TAP) per day tracks the average length of breaks between mobile sessions. Shorter breaks between mobile sessions are linked to stronger habit formation and reduced capacity to sustain attention over time [66].

Individual-week level panel. We constructed a panel at the individual-week level for each metric, spanning from 8 weeks prior to 9 weeks post-adoption. For this, we transformed calendar time into time relative to the adoption event in our panels. All mobile usage metrics were first calculated at the individual-day level and then aggregated to the week level. Specifically, we computed the

weekly sum for total duration and total number of sessions, and the weekly average for TAP. Aggregation reduces day-to-day variation while preserving the overall dynamics of mobile usage prior to and post the adoption event.

Identifying individual-level nighttime. To determine individual-level nighttime periods, we constructed a 10-hour window per user based on their typical longest period of mobile inactivity (as a proxy for sleep). For each individual and day, we identified the longest non-session interval and computed its midpoint. We then aggregated across days to obtain the median midpoint for each individual, which we treated as the center of their typical sleep period. A symmetric 10-hour window (± 5 hours around this midpoint) was defined as the nighttime for that user, with the remaining hours classified as daytime. We also analyzed effects with an 8-hour and 6-hour nighttime to ensure that our results are not driven by the size of the window (see Supplementary Tables S4 and S5).

4.2 Empirical model

4.2.1 Identification strategy

For our identification strategy, we carefully constructed comparable treatment and control groups to estimate the effect of short-form video platforms. Specifically, we aimed to find comparable groups that primarily differed in whether they were exposed to short-form videos. Hence, we defined adopters of TikTok as the treatment group and adopters of Instagram or Facebook as our control group, during a period before the latter introduced their own short-form video format. This ensures that both groups adopt platforms that are largely comparable but mainly differ in their exposure to short-form videos. Our identification relies on exploiting the adoption event under the parallel trends assumption, i.e., that, in the absence of short-form video platforms, the treatment and control groups would have evolved similarly over time. Specifically, we compared changes in mobile usage before and after adoption for individuals using short-form video platforms

(TikTok group) with those for individuals using general social media platforms without short-form videos (Instagram/Facebook group). Importantly, the parallel trends assumption is a key strength of DiD frameworks as it replaces the strong assumption of “no unobserved confounders” within observational studies, as long as those confounders are time-invariant. We further performed several robustness checks, suggesting that the estimated effects are primarily driven by the impact of exposure to short-form videos rather than the general effect of adopting a social media app (see Supplement C.2).

Our empirical model addressed several identification challenges in estimating the causal effect of short-form video platforms on mobile usage. First, individuals in our panel adopted the respective social media application at different times, which introduces a staggered adoption design. If not accounted for, this could lead to biased effect estimates due to problematic comparisons (i.e., as period-specific shocks such as holidays or exam seasons systematically affect mobile use). We addressed this by using a stacked, dynamic difference-in-differences (DiD) design that aligns treatment and control individuals by their adoption week to so-called subexperiments. We also addressed the challenge that individuals from the treatment and control groups may systematically differ by applying propensity score weighting based on pre-adoption mobile usage. Finally, to capture not only average effects but also their dynamics over time, we employ a leads-and-lags specification that provides dynamic effect estimates for each week relative to the adoption date. Detailed explanations of both the stacked design as well as the propensity score weighting are below.

Formally, our estimand of interest is the average treatment effect on the treated (ATT) at relative time l since adoption, denoted $ATT(l)$. Let $Y_{it}(1)$ denote the mobile usage outcome for individual i in week t if the individual was using short-form video platforms (treatment), and $Y_{it}(0)$ the counterfactual outcome if the individual instead was not using short-form video platforms (control).

Then, for treated individuals ($T_i = 1$), we define

$$\text{ATT}(l) = \mathbb{E}[Y_{i,t}(1) - Y_{i,t}(0) \mid T_i = 1, t - g_i = l], \quad (1)$$

where g_i is the adoption week of individual i , and l indexes time relative to adoption. We estimate $\text{ATT}(l)$ separately for three outcome metrics that capture different dimensions of mobile usage: (i) daily mobile usage duration, (ii) the number of mobile sessions, and (iii) average time away from mobile phone (TAP).

4.2.2 Stacked, dynamic DiD

We used a stacked, dynamic DiD design [37, 38] to estimate the effect of short-form video platforms $\text{ATT}(l)$. This design has several advantages compared with the standard two-way fixed effect specification. First, our stacked, dynamic DiD design stacks individuals from the treatment and control groups into so-called subexperiments based on the calendar week of adoption and therefore avoids comparisons when adoption timings vary, which can bias the estimation [67]. Moreover, our framework enables us to estimate not only the overall treatment effect but also the dynamic treatment effect based on the relative duration of short-form video platforms. Importantly, the pre-adoption effect estimates provide a pre-test of the parallel trends assumption (see Supplement C.1 for more details), while the post-adoption effect estimates reveal the trajectory of treatment effects over time.

Formally, we estimated the following model for each mobile usage metric:

$$\log(Y_{i,t,d}) = \sum_{j \in L} \beta_j D_{t,d}^j + \sum_{l \in L} \delta_l (T_{i,d} \times D_{t,d}^l) + \theta_{id} + \varepsilon_{i,t,d}, \quad (2)$$

where $Y_{i,t,d}$ is the outcome (i.e., the mobile usage metric) for individual i at time t within subexperiment d , L denotes the set of relative weeks included in our specification³, β_j captures the

³In our case, L includes the eight weeks before and nine weeks after adoption, except the omitted periods -1 and

relative week fixed effects, $D_{t,d}^j$ is an indicator variable for individuals in subexperiment d being j weeks away from adopting, δ_l is the effect estimate of short-form video platforms, and $\theta_{i,d}$ is the individual-subexperiment-level fixed effect. Due to the leads-and-lags specification, we obtained coefficient estimates that allowed us to estimate the effect of short-form video platforms $\{\delta_l\}_{l \geq 1}$ and to conduct the pre-test for the plausibility of the parallel trend assumption $\{\delta_l\}_{l < 1}$ (see Supplement C.1 for more details on the pre-test). We note that, in the special case where each individual is only in one subexperiment, $\theta_{i,d}$ simplifies to a unit-level fixed effect.

Each mobile usage metric has a specific underlying distribution (see Supplementary Figure S1), because of which we used different estimators. (1) The total duration of mobile use and (2) total number of mobile sessions were right-skewed and non-negative count variables. Therefore, we opted for a Poisson regression to ensure strictly positive outcomes. (3) The average TAP was approximately log-normally distributed, and hence we used ordinary least squares (OLS) with a log transformation. Importantly, because all models used a log link, the estimated coefficients can be interpreted as percentage changes relative to the baseline periods: being l weeks away from adopting ($D_{t,d}^l = 1$) lead to a $(\exp(\delta_l) - 1) \times 100$ % change in the corresponding mobile usage metric. For estimation, we used the function `feglm` in the library `fixest` from R (version 4.2.1).

4.2.3 Propensity score weighting

To further ensure comparability between treatment and control groups, we applied propensity score weighting [35, 36]. Propensity scores were estimated using a probit model with treatment status as the outcome and pre-adoption mobile usage as covariates. Specifically, we computed weekly averages of each mobile usage metric in the pre-adoption period and used these to predict the likelihood of receiving the treatment p_i (i.e., adopting TikTok instead of Instagram/Facebook). Because we estimated the ATT, individuals in the control group were weighted by the odds of treatment (i.e., $\frac{p_i}{1-p_i}$), while individuals in the treatment group retained a weight of one. After

0 to avoid multicollinearity and serve as baselines.

weighting, standardized mean differences between groups fell below 0.1 across all covariates (see Supplementary Figures S3 to S5), indicating adequate balance. Because weighting rather than one-to-one matching was used, no individuals were excluded, thereby preserving the sample size and statistical power. We used the library `MatchIt` from R (version 4.2.1) for propensity score weighting.

4.2.4 Aggregation

For some analyses, we reported aggregated effects for pre- and post-adoption for simplicity. Specifically, we aggregated $\{\delta_l\}_{l < 1}$ to δ_{pre} and $\{\delta_l\}_{l \geq 1}$ to δ_{post} . We used a simple average as proposed in [43] and thus computed

$$\hat{\delta}_{pre} = \frac{1}{|L_{pre}|} \sum_{\ell \in L_{pre}} \hat{\delta}_{\ell}, \quad (3)$$

$$\hat{\delta}_{post} = \frac{1}{|L_{post}|} \sum_{\ell \in L_{post}} \hat{\delta}_{\ell}, \quad (4)$$

where L_{pre} and L_{post} denote the sets of pre- and post-adoption periods. Standard errors for the aggregated effects are obtained by applying the same linear transformation to the full variance–covariance matrix of the dynamic effect estimates.

4.2.5 Robustness checks

We further performed several robustness checks to confirm that the estimated effects are not sensitive to our specific model specification. Therefore, we compared it to a standard two-way fixed effects DiD and an interaction-weighted estimator [43] (see Supplement C.3). The results show that the effect estimates are not sensitive to the choice of the model specification and hence corroborate our findings.

4.3 Subgroup analysis

We also analyzed whether certain sociodemographic subgroups show particularly strong responses to short-form video platforms, which helps identify who is most at risk of increased mobile usage. For this purpose, we used sociodemographic variables provided by the media analytics company for each individual. As some individuals in the panel opted out of reporting sociodemographics, we discarded 304 from this analysis ($\sim 17\%$ of our panel). We applied the same stacked, dynamic DiD framework, but now on the subgroup-level. To obtain uncertainty estimates, we implemented bootstrapping with $b = 2,000$ replications and used the resulting standard errors to perform t -tests on the differences across subgroups. To account for multiple comparisons, p -values from pairwise subgroup tests were adjusted using the Holm–Bonferroni procedure [45], which controls the probability of making any false rejections within each covariate-by-outcome panel.

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Author contributions

A.M. performed the analysis. A.M. and M.H.R. designed the analysis. A.M. prepared the first draft of the manuscript. All authors contributed to conceptualization, manuscript writing, and approved the manuscript.

Competing interests

The authors declare no competing interests.

Supplements

700 **Supplement A Supplementary figures**

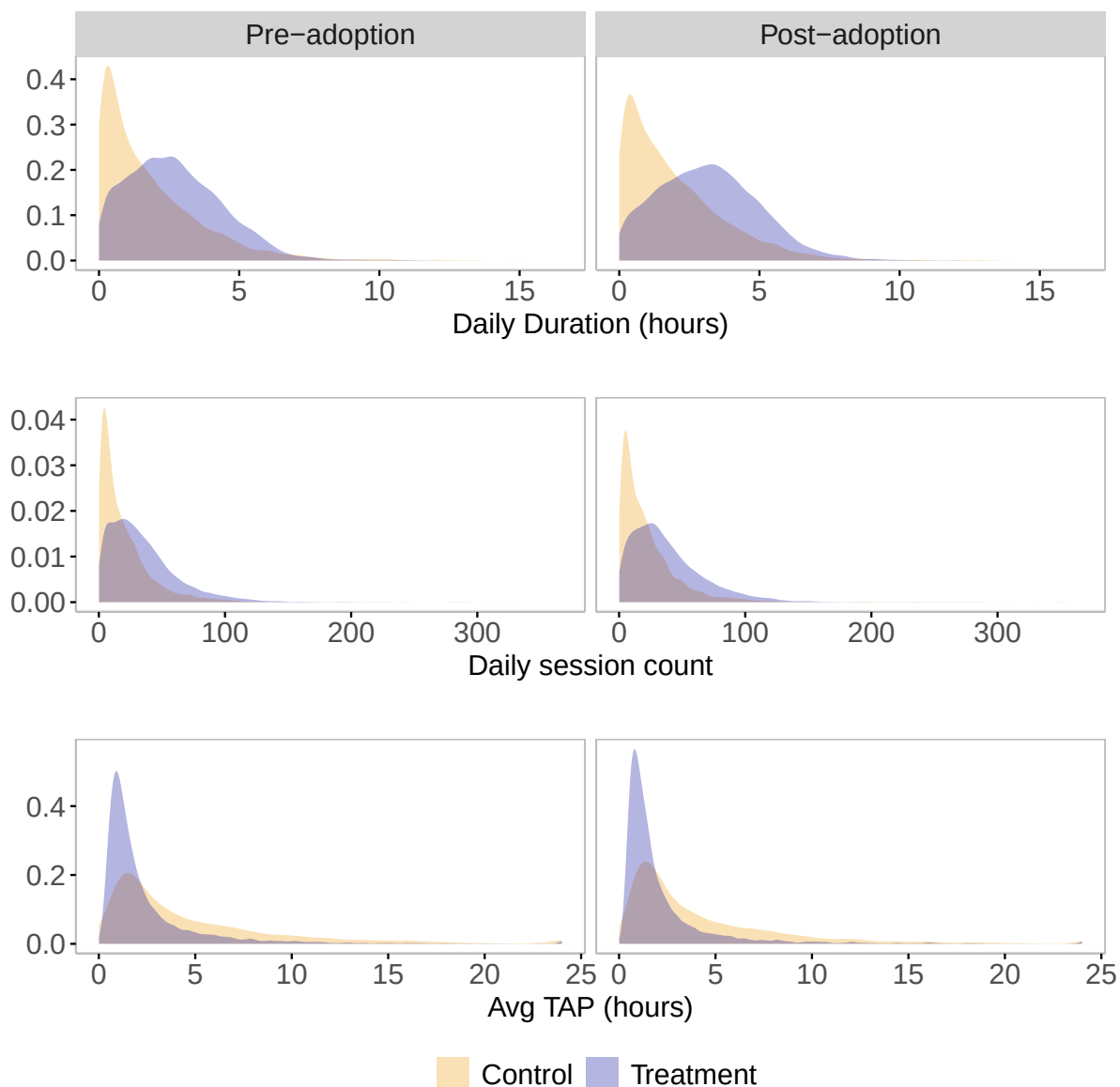


Figure S1: Distributions of the three mobile usage metrics for the treatment and control group and for pre- and post-adoption periods. To address differences between the treatment and control groups, we applied propensity score weighting based on pre-adoption mobile usage before estimating the effects of short-form video platforms (see Section 4).

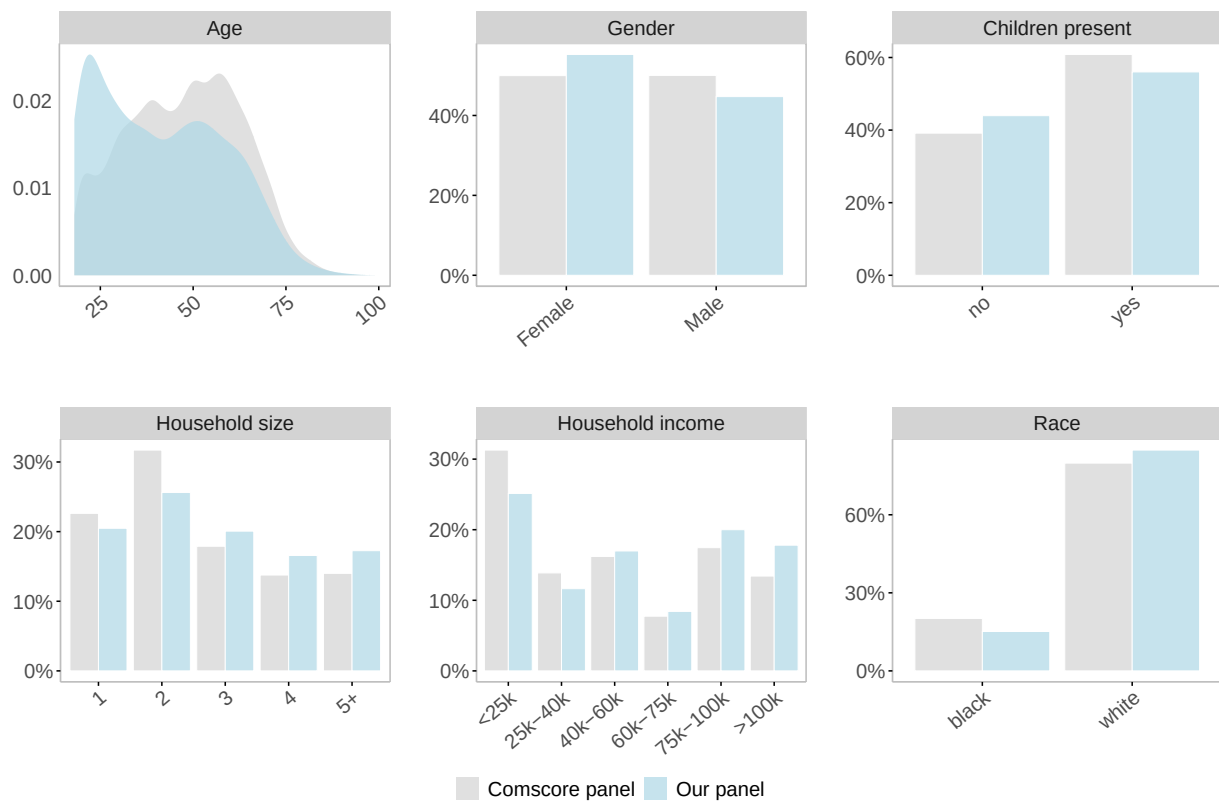


Figure S2: Comparison of the distributions of sociodemographic variables among individuals in our filtered panel vs. the entire Comscore panel.

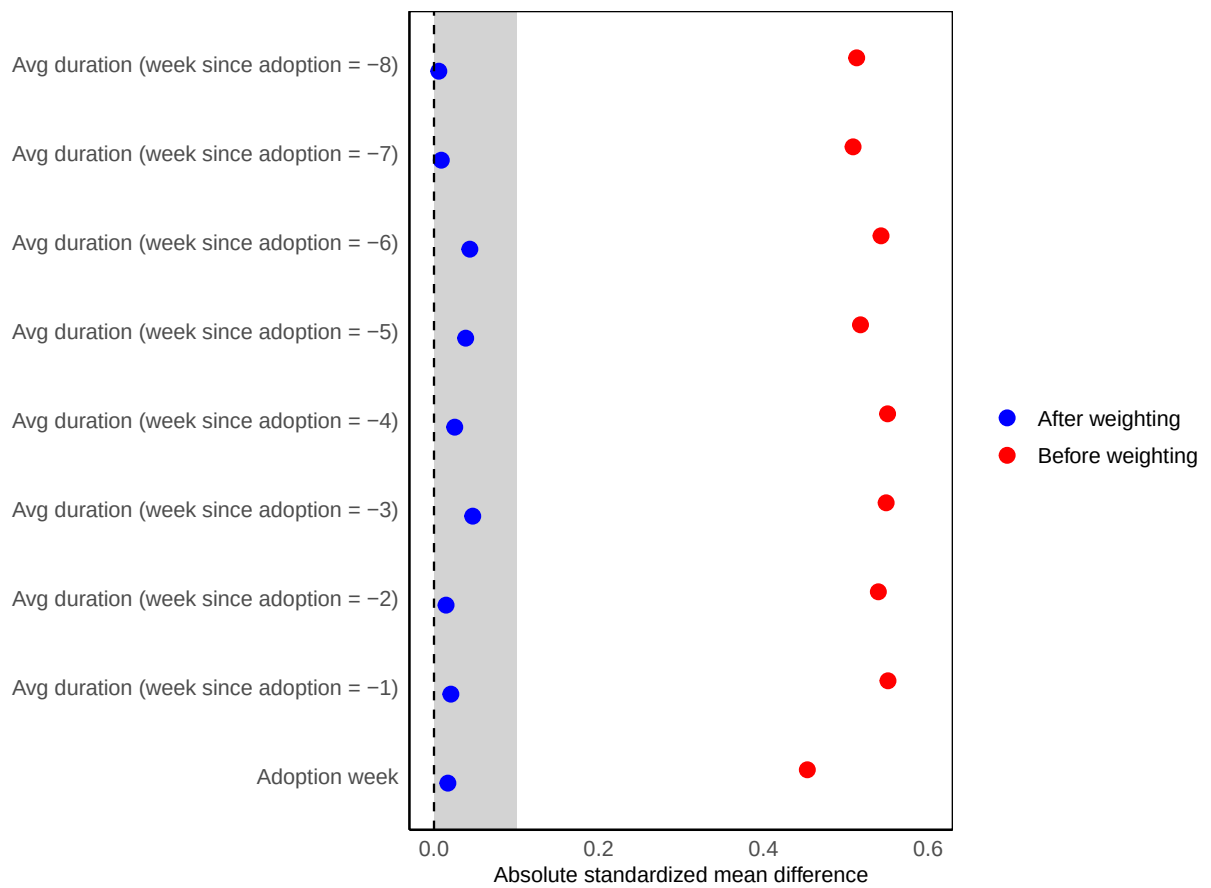


Figure S3: Absolute standardized mean differences of the variables used for propensity score weighting for the panel with the outcome of duration.

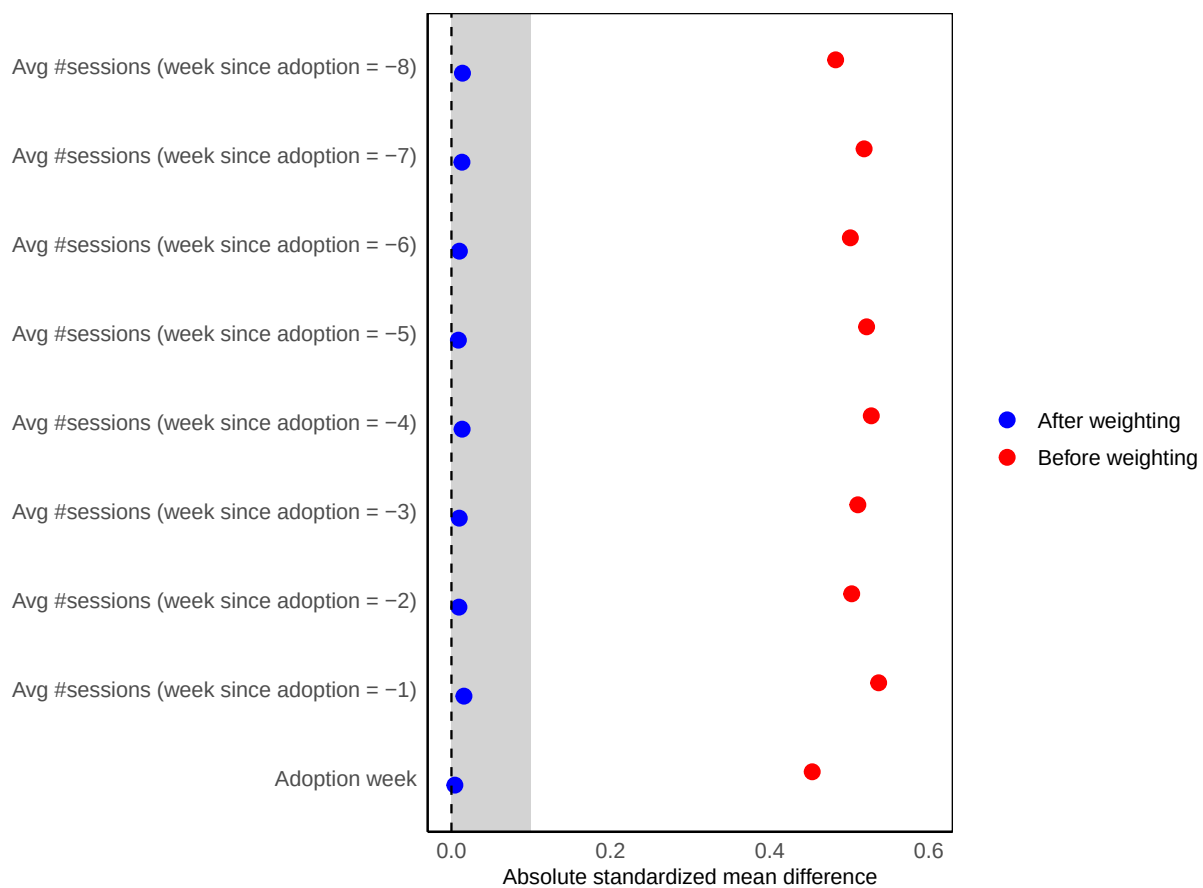


Figure S4: **Absolute standardized mean differences of the variables used for propensity score weighting for the panel with the outcome of session count.**

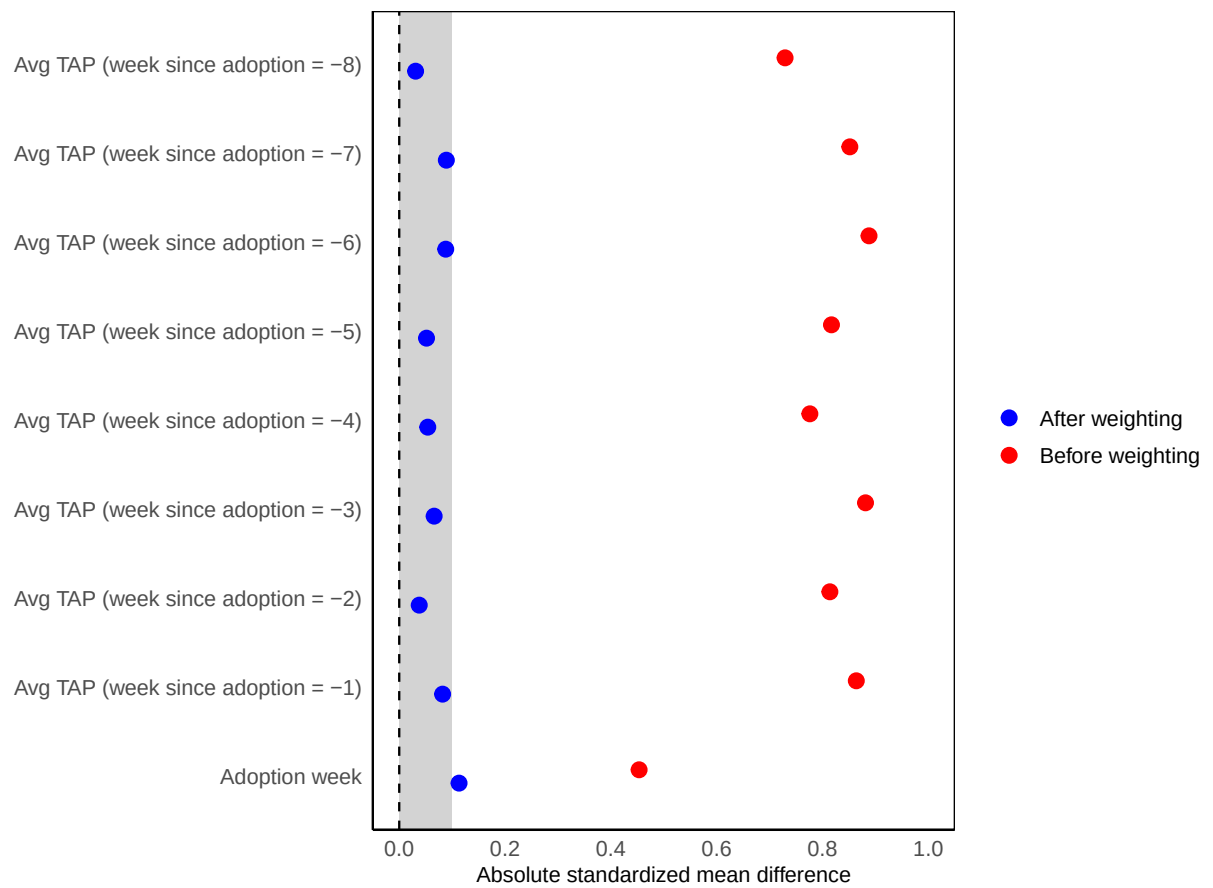


Figure S5: Absolute standardized mean differences of the variables used for propensity score weighting for the panel with the outcome of TAP.

Supplement B Supplementary tables

Dependent Variables: Distributions:	log(duration) Poisson	log(session count) Poisson	log(avg TAP) Gaussian
<i>Variables</i>			
treat × week since adoption = -8	0.007 (0.036)	0.002 (0.049)	-0.032 (0.071)
treat × week since adoption = -7	-0.002 (0.036)	-0.023 (0.048)	-0.061 (0.071)
treat × week since adoption = -6	0.024 (0.035)	-0.0010 (0.042)	-0.099 (0.070)
treat × week since adoption = -5	-0.024 (0.036)	-0.021 (0.043)	-0.042 (0.069)
treat × week since adoption = -4	-0.010 (0.037)	-0.022 (0.041)	-0.076 (0.074)
treat × week since adoption = -3	-0.025 (0.030)	-0.015 (0.044)	-0.037 (0.066)
treat × week since adoption = -2	-0.002 (0.031)	-0.017 (0.037)	-0.008 (0.068)
treat × week since adoption = 1	0.072 ** (0.035)	-0.020 (0.040)	-0.034 (0.066)
treat × week since adoption = 2	0.078 ** (0.032)	0.0003 (0.042)	-0.047 (0.068)
treat × week since adoption = 3	0.069 ** (0.033)	0.006 (0.043)	-0.135 * (0.072)
treat × week since adoption = 4	0.088 ** (0.037)	0.046 (0.046)	-0.142 * (0.073)
treat × week since adoption = 5	0.121 *** (0.040)	0.074 * (0.044)	-0.109 (0.079)
treat × week since adoption = 6	0.131 *** (0.046)	0.062 (0.050)	-0.204 ** (0.080)
treat × week since adoption = 7	0.153 *** (0.036)	0.099 ** (0.045)	-0.148 ** (0.075)
treat × week since adoption = 8	0.149 *** (0.044)	0.048 (0.049)	-0.110 (0.081)
treat × week since adoption = 9	0.161 *** (0.043)	0.079 * (0.046)	-0.220 ** (0.104)
<i>Fixed-effects</i>			
individual × subexperiment	✓	✓	✓
week since adoption	✓	✓	✓
<i>Fit statistics</i>			
Pseudo R^2	1.0653	1.1372	0.04939
Observations	29,637	29,637	29,586

Clustered (individual × subexperiment) standard-errors in parentheses

Signif. Codes: ***, 0.01, **, 0.05, *, 0.1

Table S1: Effect estimates of short-form video platforms on mobile usage metrics.

Dependent variable: Distribution:	log(duration) Poisson		log(session count) Poisson		log(avg TAP) Gaussian	
	Low	High	Low	High	Low	High
<i>Usage intensity</i>						
<i>Variables</i>						
treat × week since adoption = -8	0.087 (0.080)	-0.006 (0.040)	-0.032 (0.056)	0.014 (0.062)	-0.046 (0.079)	0.014 (0.140)
treat × week since adoption = -7	0.010 (0.076)	0.002 (0.041)	0.010 (0.054)	-0.032 (0.058)	-0.049 (0.079)	-0.107 (0.131)
treat × week since adoption = -6	0.047 (0.074)	0.024 (0.040)	0.013 (0.054)	-0.007 (0.050)	-0.113 (0.082)	-0.043 (0.114)
treat × week since adoption = -5	0.134* (0.076)	-0.055 (0.040)	0.035 (0.055)	-0.040 (0.051)	-0.070 (0.079)	0.070 (0.120)
treat × week since adoption = -4	0.057 (0.073)	-0.020 (0.041)	-0.006 (0.058)	-0.029 (0.049)	-0.097 (0.084)	0.011 (0.153)
treat × week since adoption = -3	0.029 (0.063)	-0.033 (0.033)	0.050 (0.054)	-0.036 (0.054)	-0.045 (0.077)	-0.004 (0.081)
treat × week since adoption = -2	0.015 (0.063)	-0.0010 (0.036)	0.031 (0.055)	-0.032 (0.044)	-0.006 (0.080)	-0.004 (0.086)
treat × week since adoption = 1	0.176*** (0.054)	0.047 (0.042)	0.088 (0.056)	-0.049 (0.048)	-0.058 (0.077)	0.058 (0.094)
treat × week since adoption = 2	0.187*** (0.059)	0.050 (0.037)	0.070 (0.055)	-0.015 (0.051)	-0.056 (0.079)	-0.014 (0.094)
treat × week since adoption = 3	0.177*** (0.062)	0.042 (0.039)	0.051 (0.055)	-0.002 (0.053)	-0.166* (0.085)	-0.019 (0.089)
treat × week since adoption = 4	0.195*** (0.069)	0.060 (0.043)	0.076 (0.058)	0.044 (0.057)	-0.173** (0.084)	-0.030 (0.112)
treat × week since adoption = 5	0.200*** (0.070)	0.102** (0.048)	0.078 (0.060)	0.081 (0.054)	-0.104 (0.087)	-0.136 (0.173)
treat × week since adoption = 6	0.261*** (0.075)	0.097* (0.055)	0.074 (0.062)	0.067 (0.063)	-0.258*** (0.092)	-0.007 (0.122)
treat × week since adoption = 7	0.246*** (0.068)	0.129*** (0.043)	0.087 (0.062)	0.114** (0.056)	-0.149* (0.087)	-0.153 (0.123)
treat × week since adoption = 8	0.235*** (0.076)	0.127** (0.053)	0.110* (0.059)	0.036 (0.062)	-0.130 (0.090)	-0.057 (0.175)
treat × week since adoption = 9	0.273*** (0.079)	0.131*** (0.051)	0.104* (0.061)	0.083 (0.057)	-0.254** (0.125)	-0.111 (0.124)
<i>Fixed-effects</i>						
individual × subexperiment	✓	✓	✓	✓	✓	✓
week since adoption	✓	✓	✓	✓	✓	✓
<i>Fit statistics</i>						
Pseudo R ²	0.87498	1.0293	0.68025	1.0634	0.04213	0.04324
Observations	14,708	14,929	14,708	14,929	14,662	14,924

Clustered (individual × subexperiment) standard-errors in parentheses
Signif. Codes: ***, 0.01, **, 0.05, *, 0.1

Table S2: Effect estimates of short-form video platforms on mobile usage metrics for low vs. high-intensity users.

Dependent variable: Distribution:	log(duration) Poisson		log(session count) Poisson		log(avg TAP) Gaussian	
	Day	Night	Day	Night	Day	Night
<i>Variables</i>						
treat × week since adoption = -8	0.029 (0.037)	-0.051 (0.075)	-0.0003 (0.055)	0.016 (0.061)	-0.087 (0.092)	0.021 (0.033)
treat × week since adoption = -7	0.020 (0.038)	-0.087 (0.078)	-0.028 (0.054)	-0.020 (0.056)	-0.149* (0.090)	0.011 (0.033)
treat × week since adoption = -6	0.025 (0.039)	0.013 (0.061)	-0.013 (0.048)	0.040 (0.052)	-0.164* (0.090)	0.000 (0.028)
treat × week since adoption = -5	-0.021 (0.038)	-0.053 (0.070)	-0.028 (0.048)	0.004 (0.046)	-0.052 (0.090)	0.0004 (0.028)
treat × week since adoption = -4	0.004 (0.041)	-0.047 (0.091)	-0.029 (0.048)	0.006 (0.046)	-0.160* (0.093)	0.017 (0.027)
treat × week since adoption = -3	-0.046 (0.033)	0.037 (0.054)	-0.043 (0.050)	0.073* (0.042)	-0.073 (0.085)	-0.021 (0.028)
treat × week since adoption = -2	0.005 (0.034)	-0.020 (0.058)	-0.025 (0.043)	0.008 (0.036)	-0.072 (0.089)	-0.008 (0.025)
treat × week since adoption = 1	0.097** (0.042)	-0.003 (0.055)	-0.011 (0.045)	-0.050 (0.040)	-0.009 (0.084)	0.029 (0.025)
treat × week since adoption = 2	0.100*** (0.036)	0.010 (0.058)	-0.014 (0.045)	0.048 (0.057)	-0.130 (0.091)	-0.012 (0.030)
treat × week since adoption = 3	0.085** (0.037)	0.023 (0.056)	-0.005 (0.046)	0.038 (0.052)	-0.119 (0.089)	-0.004 (0.028)
treat × week since adoption = 4	0.096** (0.038)	0.061 (0.063)	0.054 (0.050)	0.017 (0.051)	-0.239** (0.094)	-0.024 (0.027)
treat × week since adoption = 5	0.119*** (0.040)	0.131* (0.073)	0.058 (0.046)	0.130** (0.055)	-0.075 (0.096)	0.003 (0.027)
treat × week since adoption = 6	0.130** (0.050)	0.129* (0.066)	0.056 (0.050)	0.063 (0.066)	-0.193** (0.086)	0.005 (0.029)
treat × week since adoption = 7	0.161*** (0.040)	0.128** (0.060)	0.104** (0.047)	0.071 (0.060)	-0.191** (0.095)	-0.007 (0.033)
treat × week since adoption = 8	0.176*** (0.043)	0.074 (0.079)	0.044 (0.052)	0.047 (0.061)	-0.115 (0.097)	0.027 (0.029)
treat × week since adoption = 9	0.174*** (0.042)	0.091 (0.078)	0.093** (0.045)	-0.002 (0.061)	-0.327** (0.143)	-0.020 (0.029)
<i>Fixed-effects</i>						
individual × subexperiment	✓	✓	✓	✓	✓	✓
week since adoption	✓	✓	✓	✓	✓	✓
<i>Fit statistics</i>						
Pseudo R ²	1.0707	1.2375	1.1644	2.3150	0.04388	1.6132
Observations	29,339	28,607	29,339	28,607	29,328	28,654

Clustered (individual × subexperiment) standard-errors in parentheses
Signif. Codes: ***, 0.01, **, 0.05, *, 0.1

Table S3: Effect estimates of short-form video platforms on mobile usage metrics for day vs. nighttime.

Dependent variable: Distribution:	log(duration) Poisson		log(session count) Poisson		log(avg TAP) Gaussian	
	Day	Night	Day	Night	Day	Night
<i>Variables</i>						
treat × week since adoption = -8	0.025 (0.037)	-0.073 (0.095)	-0.006 (0.053)	0.049 (0.082)	-0.044 (0.089)	0.004 (0.042)
treat × week since adoption = -7	0.015 (0.037)	-0.103 (0.103)	-0.034 (0.051)	0.027 (0.076)	-0.105 (0.091)	-0.011 (0.041)
treat × week since adoption = -6	0.025 (0.036)	0.002 (0.069)	-0.013 (0.045)	0.059 (0.066)	-0.136 (0.089)	-0.007 (0.042)
treat × week since adoption = -5	-0.018 (0.035)	-0.102 (0.083)	-0.039 (0.045)	0.089 (0.058)	-0.029 (0.086)	0.029 (0.041)
treat × week since adoption = -4	0.011 (0.038)	-0.091 (0.114)	-0.035 (0.045)	0.057 (0.055)	-0.107 (0.091)	0.007 (0.035)
treat × week since adoption = -3	-0.036 (0.031)	0.047 (0.072)	-0.034 (0.048)	0.103** (0.048)	-0.056 (0.086)	0.003 (0.043)
treat × week since adoption = -2	0.004 (0.033)	-0.035 (0.073)	-0.019 (0.042)	-0.009 (0.043)	-0.062 (0.087)	-0.016 (0.036)
treat × week since adoption = 1	0.088** (0.040)	0.004 (0.062)	-0.012 (0.042)	-0.068 (0.050)	-0.010 (0.084)	0.023 (0.036)
treat × week since adoption = 2	0.094*** (0.034)	0.001 (0.076)	-0.008 (0.044)	0.050 (0.071)	-0.107 (0.087)	-0.056 (0.041)
treat × week since adoption = 3	0.082** (0.036)	0.008 (0.075)	-0.003 (0.044)	0.049 (0.064)	-0.142 (0.095)	-0.010 (0.043)
treat × week since adoption = 4	0.086** (0.038)	0.098 (0.081)	0.050 (0.048)	0.014 (0.061)	-0.228** (0.089)	-0.037 (0.038)
treat × week since adoption = 5	0.124*** (0.041)	0.108 (0.081)	0.061 (0.046)	0.148** (0.063)	-0.049 (0.095)	0.003 (0.044)
treat × week since adoption = 6	0.129*** (0.048)	0.139* (0.078)	0.049 (0.049)	0.119 (0.082)	-0.162* (0.085)	-0.025 (0.044)
treat × week since adoption = 7	0.157*** (0.038)	0.141** (0.068)	0.099** (0.046)	0.091 (0.073)	-0.190** (0.092)	-0.058 (0.045)
treat × week since adoption = 8	0.159*** (0.044)	0.100 (0.090)	0.038 (0.050)	0.075 (0.076)	-0.095 (0.102)	0.032 (0.042)
treat × week since adoption = 9	0.175*** (0.042)	0.063 (0.101)	0.090** (0.045)	0.006 (0.073)	-0.351** (0.140)	-0.004 (0.043)
<i>Fixed-effects</i>						
individual × subexperiment	✓	✓	✓	✓	✓	✓
week since adoption	✓	✓	✓	✓	✓	✓
<i>Fit statistics</i>						
Pseudo R ²	1.0673	1.4746	1.1533	-4.9679	0.04472	0.44959
Observations	29,451	27,367	29,451	27,367	29,418	27,558

Clustered (individual × subexperiment) standard-errors in parentheses
Signif. Codes: ***, 0.01, **, 0.05, *, 0.1

Table S4: Effect estimates of short-form video platforms on mobile usage metrics for day vs. nighttime with an 8-hour night window.

Dependent variable: Distribution:	log(duration) Poisson		log(session count) Poisson		log(avg TAP) Gaussian	
	Day	Night	Day	Night	Day	Night
<i>Variables</i>						
treat × week since adoption = -8	0.022 (0.036)	-0.125 (0.100)	-0.005 (0.051)	0.087 (0.104)	-0.057 (0.081)	-0.001 (0.067)
treat × week since adoption = -7	0.015 (0.035)	-0.152 (0.101)	-0.031 (0.049)	0.090 (0.102)	-0.138* (0.083)	-0.015 (0.056)
treat × week since adoption = -6	0.031 (0.036)	-0.086 (0.100)	-0.008 (0.043)	0.066 (0.096)	-0.140* (0.081)	-0.044 (0.062)
treat × week since adoption = -5	-0.021 (0.035)	-0.124 (0.097)	-0.037 (0.044)	0.148* (0.087)	-0.047 (0.081)	-0.006 (0.055)
treat × week since adoption = -4	0.006 (0.038)	-0.183 (0.126)	-0.033 (0.043)	0.090 (0.077)	-0.105 (0.083)	-0.050 (0.050)
treat × week since adoption = -3	-0.027 (0.030)	6.22×10^{-5} (0.087)	-0.024 (0.046)	0.085 (0.063)	-0.041 (0.075)	-0.008 (0.051)
treat × week since adoption = -2	0.001 (0.032)	-0.092 (0.090)	-0.021 (0.040)	0.021 (0.057)	-0.038 (0.078)	-0.017 (0.054)
treat × week since adoption = 1	0.085** (0.040)	-0.052 (0.088)	-0.011 (0.041)	-0.127** (0.059)	-0.013 (0.077)	-0.013 (0.057)
treat × week since adoption = 2	0.092*** (0.034)	-0.078 (0.099)	-0.008 (0.043)	0.092 (0.089)	-0.056 (0.077)	-0.063 (0.058)
treat × week since adoption = 3	0.077** (0.036)	-0.013 (0.094)	8.11×10^{-6} (0.044)	0.060 (0.076)	-0.166** (0.081)	0.043 (0.059)
treat × week since adoption = 4	0.092** (0.037)	0.035 (0.106)	0.055 (0.046)	-0.062 (0.076)	-0.170** (0.081)	-0.0009 (0.055)
treat × week since adoption = 5	0.124*** (0.041)	0.053 (0.098)	0.064 (0.045)	0.182** (0.076)	-0.088 (0.086)	0.065 (0.060)
treat × week since adoption = 6	0.128*** (0.046)	0.098 (0.101)	0.056 (0.048)	0.117 (0.106)	-0.148* (0.078)	-0.025 (0.058)
treat × week since adoption = 7	0.154*** (0.038)	0.112 (0.092)	0.096** (0.046)	0.143 (0.094)	-0.184** (0.084)	-0.012 (0.061)
treat × week since adoption = 8	0.157*** (0.044)	0.037 (0.107)	0.049 (0.049)	0.013 (0.087)	-0.139 (0.091)	0.042 (0.057)
treat × week since adoption = 9	0.168*** (0.042)	0.010 (0.146)	0.086* (0.044)	-0.028 (0.087)	-0.330** (0.131)	-0.009 (0.057)
<i>Fixed-effects</i>						
individual × subexperiment	✓	✓	✓	✓	✓	✓
week since adoption	✓	✓	✓	✓	✓	✓
<i>Fit statistics</i>						
Pseudo R ²	1.0662	2.1058	1.1466	-0.40585	0.04564	0.78040
Observations	29,519	24,762	29,519	24,762	29,492	25,206

Clustered (individual × subexperiment) standard-errors in parentheses
Signif. Codes: ***, 0.01, **, 0.05, *, 0.1

Table S5: Effect estimates of short-form video platforms on mobile usage metrics for day vs. nighttime with a 6-hour night window.

Dependent variables:	Duration	Session count	Avg TAP	Obs
<i>Age</i>				
young	9.476 (4.580)	-0.428 (3.514)	-17.439 (10.905)	307
middle	14.123 (4.479)	-5.606 (3.419)	-8.095 (8.277)	500
old	11.842 (4.115)	9.551 (3.144)	-8.254 (7.603)	631
<i>Gender</i>				
male	7.941 (3.554)	7.294 (2.823)	-10.732 (7.023)	644
female	15.306 (3.600)	-1.154 (2.917)	-8.738 (6.385)	794
<i>Race</i>				
white	11.491 (2.721)	1.897 (2.028)	-10.324 (5.346)	1,221
black	12.422 (6.666)	14.808 (6.048)	-4.473 (13.797)	217
<i>Children present</i>				
yes	8.351 (3.526)	-1.509 (2.564)	-20.931 (5.732)	629
no	14.913 (3.572)	7.047 (2.881)	1.774 (7.396)	809
<i>Household income</i>				
<25k	11.981 (4.105)	4.724 (3.584)	-7.613 (8.876)	361
25k-40k	24.341 (9.000)	11.480 (8.693)	-1.580 (18.947)	168
40k-60k	14.450 (4.781)	2.029 (4.748)	3.082 (13.660)	245
60k-75k	-4.102 (9.482)	3.530 (7.027)	-16.386 (11.932)	120
75k-100k	13.569 (7.299)	-1.494 (3.436)	6.424 (10.798)	288
>100k	12.579 (7.347)	7.443 (4.813)	-20.555 (10.062)	257
<i>Household size</i>				
1	6.521 (5.332)	-7.050 (4.306)	0.712 (12.549)	295
2	20.317 (5.768)	12.285 (4.214)	11.464 (10.903)	371
3	29.861 (5.153)	3.867 (3.826)	-10.938 (9.544)	286
4	1.642 (4.652)	7.323 (3.550)	-30.253 (7.242)	236
5+	1.582 (8.648)	-7.665 (4.990)	-9.909 (13.438)	250

Table S6: **Aggregated effect estimates (in %) of short-form video platforms on mobile usage metrics per sociodemographic subgroup.** Here, we report the aggregated effect estimates post-adoption for each sociodemographic subgroup using the aggregation method from [43]. To obtain uncertainty estimates, we used bootstrapping with $b = 2000$ replications and used the resulting standard errors to perform t -tests for statistically significant differences across subgroups, as reported in Figure 4.

Dependent variables:	Duration	Session count	Avg TAP	Obs
<i>Age</i>				
young	3.065 (0.033)	35.340 (0.467)	2.101 (0.050)	307
middle	2.700 (0.028)	35.933 (0.507)	2.660 (0.052)	500
old	1.919 (0.024)	24.047 (0.361)	4.013 (0.058)	631
<i>Gender</i>				
male	2.212 (0.026)	25.425 (0.330)	3.642 (0.055)	644
female	2.617 (0.022)	34.762 (0.385)	2.721 (0.042)	794
<i>Race</i>				
white	2.400 (0.016)	29.543 (0.248)	3.219 (0.034)	1,521
black	2.515 (0.046)	31.257 (0.663)	3.098 (0.087)	217
<i>Children present</i>				
yes	2.633 (0.024)	32.901 (0.399)	2.774 (0.047)	629
no	2.279 (0.023)	28.760 (0.347)	3.418 (0.048)	809
<i>Household income</i>				
<25k	2.669 (0.034)	30.489 (0.532)	2.895 (0.063)	361
25k-40k	2.416 (0.048)	27.852 (0.646)	3.196 (0.102)	168
40k-60k	2.416 (0.041)	32.666 (0.642)	3.018 (0.079)	245
60k-75k	2.152 (0.050)	28.946 (0.855)	3.251 (0.113)	120
75k-100k	2.407 (0.037)	30.622 (0.593)	3.204 (0.077)	288
>100k	2.311 (0.042)	31.383 (0.667)	3.395 (0.087)	257
<i>Household size</i>				
1	2.339 (0.038)	27.982 (0.536)	3.488 (0.079)	295
2	2.221 (0.034)	28.962 (0.546)	3.447 (0.071)	371
3	2.425 (0.035)	31.748 (0.555)	2.874 (0.069)	286
4	2.601 (0.040)	31.385 (0.618)	2.944 (0.081)	236
5+	2.728 (0.040)	34.035 (0.690)	2.729 (0.077)	250

Table S7: Comparison of the mean (and standard error) of each mobile usage metric across different sociodemographic subgroups prior adoption.

Dependent Variables: Distributions:	log(duration) Poisson	log(session count) Poisson	log(avg TAP) Gaussian
<i>Variables</i>			
treat × week since adoption = -8	-0.084** (0.039)	0.040 (0.047)	-0.040 (0.083)
treat × week since adoption = -7	-0.106** (0.042)	0.025 (0.052)	-0.005 (0.070)
treat × week since adoption = -6	-0.063* (0.037)	0.036 (0.038)	-0.041 (0.077)
treat × week since adoption = -5	-0.049 (0.042)	0.027 (0.038)	-0.006 (0.075)
treat × week since adoption = -4	0.0003 (0.038)	0.034 (0.040)	-0.061 (0.074)
treat × week since adoption = -3	0.007 (0.035)	0.001 (0.040)	-0.013 (0.068)
treat × week since adoption = -2	0.0003 (0.043)	-0.010 (0.033)	0.007 (0.072)
treat × week since adoption = 1	0.023 (0.037)	0.004 (0.037)	0.022 (0.067)
treat × week since adoption = 2	0.005 (0.035)	0.051 (0.040)	-0.016 (0.069)
treat × week since adoption = 3	0.007 (0.036)	0.058 (0.054)	-0.201 (0.135)
treat × week since adoption = 4	0.039 (0.041)	0.098** (0.046)	-0.198 (0.141)
treat × week since adoption = 5	0.133*** (0.040)	0.125** (0.050)	-0.138 (0.118)
treat × week since adoption = 6	0.081** (0.041)	0.047 (0.057)	-0.273* (0.150)
treat × week since adoption = 7	0.126*** (0.040)	0.089 (0.054)	-0.329* (0.182)
treat × week since adoption = 8	0.126*** (0.044)	0.045 (0.064)	-0.225 (0.167)
treat × week since adoption = 9	0.184** (0.079)	0.112 (0.071)	-0.196 (0.133)
<i>Fixed-effects</i>			
individual × subexperiment	✓	✓	✓
week since adoption	✓	✓	✓
<i>Fit statistics</i>			
Pseudo R ²	1.0628	1.1850	0.04774
Observations	25,227	25,227	25,185

Clustered (individual × subexperiment) standard-errors in parentheses
Signif. Codes: ***, 0.01, **, 0.05, *, 0.1

Table S8: Effect estimates of exposure to short-form videos on mobile usage metrics with an adoption window of 5 days.

Dependent Variables: Distributions:	log(duration) Poisson	log(session count) Poisson	log(avg TAP) Gaussian
<i>Variables</i>			
treat $\times$ week since adoption = -8	-0.071 (0.073)	0.067 (0.048)	0.010 (0.089)
treat $\times$ week since adoption = -7	-0.125 (0.076)	0.009 (0.052)	-0.035 (0.080)
treat $\times$ week since adoption = -6	-0.054 (0.069)	0.039 (0.046)	-0.046 (0.089)
treat $\times$ week since adoption = -5	-0.048 (0.064)	0.015 (0.053)	-0.011 (0.088)
treat $\times$ week since adoption = -4	0.010 (0.055)	0.050 (0.051)	-0.029 (0.092)
treat $\times$ week since adoption = -3	0.011 (0.047)	0.058 (0.042)	-0.031 (0.084)
treat $\times$ week since adoption = -2	-0.040 (0.043)	0.050 (0.041)	-0.024 (0.089)
treat $\times$ week since adoption = 1	0.050 (0.038)	0.044 (0.043)	0.045 (0.087)
treat $\times$ week since adoption = 2	0.075 (0.049)	0.094* (0.045)	-0.023 (0.085)
treat $\times$ week since adoption = 3	0.083* (0.043)	0.091* (0.048)	-0.241* (0.124)
treat $\times$ week since adoption = 4	0.127*** (0.048)	0.125** (0.051)	-0.226* (0.131)
treat $\times$ week since adoption = 5	0.159*** (0.041)	0.125*** (0.047)	-0.112 (0.118)
treat $\times$ week since adoption = 6	0.183*** (0.047)	0.139** (0.057)	-0.186 (0.140)
treat $\times$ week since adoption = 7	0.141*** (0.046)	0.154*** (0.054)	-0.233 (0.162)
treat $\times$ week since adoption = 8	0.192*** (0.043)	0.100* (0.052)	-0.179 (0.162)
treat $\times$ week since adoption = 9	0.185*** (0.049)	0.144*** (0.053)	-0.239* (0.124)
treat $\times$ week since adoption = 11	0.128* (0.066)	0.111* (0.060)	-0.250* (0.129)
treat $\times$ week since adoption = 12	0.103 (0.152)	0.195* (0.118)	0.018 (0.138)
<i>Fixed-effects</i>			
individual $\times$ subexperiment	✓	✓	✓
week since adoption	✓	✓	✓
<i>Fit statistics</i>			
Pseudo R ²	1.0775	1.2081	0.04655
Observations	24,330	24,330	24,290

Clustered (individual $\times$ subexperiment) standard-errors in parentheses
Signif. Codes: ***, 0.01, **, 0.05, *, 0.1

Table S9: Effect estimates of short-form video platforms on mobile usage metrics with an adoption window of 20 days.

Dependent Variables: Distributions:	log(duration) Poisson	log(session count) Poisson	log(avg TAP) Gaussian
<i>Variables</i>			
treat $\times$ week since adoption = -8	-0.034 (0.026)	-0.009 (0.021)	0.014 (0.032)
treat $\times$ week since adoption = -7	-0.040 (0.026)	0.005 (0.023)	0.025 (0.029)
treat $\times$ week since adoption = -6	-0.019 (0.024)	-0.003 (0.020)	0.047 (0.030)
treat $\times$ week since adoption = -5	-0.003 (0.022)	-0.014 (0.022)	0.021 (0.031)
treat $\times$ week since adoption = -4	0.011 (0.022)	-0.023 (0.023)	0.028 (0.030)
treat $\times$ week since adoption = -3	-0.004 (0.022)	-0.019 (0.022)	0.037 (0.029)
treat $\times$ week since adoption = -2	-0.007 (0.021)	-0.014 (0.021)	0.021 (0.038)
treat $\times$ week since adoption = 1	0.056*** (0.021)	0.007 (0.021)	-0.018 (0.033)
treat $\times$ week since adoption = 2	0.038* (0.022)	0.012 (0.021)	-0.028 (0.032)
treat $\times$ week since adoption = 3	0.035* (0.021)	-0.007 (0.022)	-0.025 (0.034)
treat $\times$ week since adoption = 4	0.019 (0.020)	0.005 (0.021)	-0.007 (0.037)
treat $\times$ week since adoption = 5	0.058*** (0.021)	0.011 (0.022)	-0.026 (0.032)
treat $\times$ week since adoption = 6	0.067*** (0.023)	0.024 (0.022)	-0.026 (0.033)
treat $\times$ week since adoption = 7	0.088*** (0.025)	0.026 (0.026)	-0.013 (0.034)
treat $\times$ week since adoption = 8	0.085*** (0.024)	-0.018 (0.027)	0.018 (0.033)
treat $\times$ week since adoption = 9	0.088*** (0.027)	6.95×10^{-6} (0.023)	-0.007 (0.034)
<i>Fixed-effects</i>			
individual $\times$ subexperiment	✓	✓	✓
week since adoption	✓	✓	✓
<i>Fit statistics</i>			
Pseudo R ²	1.0712	1.0921	0.05578
Observations	98,940	98,940	98,718

Clustered (individual $\times$ subexperiment) standard-errors in parentheses

Signif. Codes: ***, 0.01, **, 0.05, *, 0.1

Table S10: Effect estimates of short-form video platforms on mobile usage metrics with no adoption window. Here, we report the effect estimates with no adoption window. Hence, individuals who used an application for the first time are regarded as adopters. We note that this definition of adoption may be weak, as it can include individuals who only installed the application and never used it again. Therefore, individuals from the treatment and control group may not differ in terms of mobile usage post-adoption, as most individuals in the treatment group who should have used short-form video platforms, as per our research question, have perhaps not used them.

Dependent Variables: Distributions:	log(duration) Poisson	log(session count) Poisson	log(avg TAP) Gaussian
<i>Variables</i>			
treat $\times$ week since adoption = -5	-0.053 (0.091)	-0.126 (0.107)	0.060 (0.171)
treat $\times$ week since adoption = -4	-0.020 (0.039)	0.061 (0.061)	0.026 (0.066)
treat $\times$ week since adoption = -3	-0.025 (0.029)	0.024 (0.031)	-0.008 (0.059)
treat $\times$ week since adoption = -2	-0.025 (0.026)	-0.006 (0.021)	0.045 (0.065)
treat $\times$ week since adoption = 1	0.057** (0.027)	0.013 (0.025)	0.015 (0.072)
treat $\times$ week since adoption = 2	0.076** (0.030)	0.023 (0.027)	-0.103* (0.061)
treat $\times$ week since adoption = 3	0.088*** (0.031)	0.063* (0.037)	-0.175** (0.075)
treat $\times$ week since adoption = 4	0.092*** (0.031)	0.114*** (0.036)	-0.131* (0.073)
treat $\times$ week since adoption = 5	0.120*** (0.035)	0.142*** (0.040)	-0.176** (0.078)
treat $\times$ week since adoption = 6	0.099* (0.057)	-0.027 (0.111)	-0.287*** (0.093)
<i>Fixed-effects</i>			
individual $\times$ subexperiment	✓	✓	✓
week since adoption	✓	✓	✓
<i>Fit statistics</i>			
Pseudo R ²	1.0456	1.0885	0.04886
Observations	27,415	27,415	27,332

Clustered (individual $\times$ subexperiment) standard-errors in parentheses
Signif. Codes: ***, 0.01, **, 0.05, *, 0.1

Table S11: Effect estimates of short-form video platforms on mobile usage metrics with a period length of 30 days prior to and post the adoption window.

Dependent Variables: Distributions:	log(duration) Poisson	log(session count) Poisson	log(avg TAP) Gaussian
<i>Variables</i>			
treat × week since adoption = -12	-0.033 (0.052)	-0.024 (0.044)	-0.096 (0.085)
treat × week since adoption = -11	-0.043 (0.047)	-0.025 (0.044)	-0.079 (0.081)
treat × week since adoption = -10	-0.019 (0.048)	0.013 (0.051)	-0.102 (0.091)
treat × week since adoption = -9	0.004 (0.052)	-0.013 (0.047)	-0.142* (0.076)
treat × week since adoption = -8	-0.017 (0.045)	-0.0008 (0.044)	-0.127 (0.096)
treat × week since adoption = -7	-0.044 (0.049)	-0.019 (0.043)	-0.058 (0.067)
treat × week since adoption = -6	-0.028 (0.053)	0.034 (0.046)	-0.085 (0.083)
treat × week since adoption = -5	-0.026 (0.042)	0.030 (0.043)	-0.108 (0.079)
treat × week since adoption = -4	-0.022 (0.042)	0.019 (0.041)	-0.129* (0.076)
treat × week since adoption = -3	-0.058 (0.046)	0.011 (0.040)	-0.117 (0.083)
treat × week since adoption = -2	-0.032 (0.038)	-0.022 (0.035)	-0.117 (0.090)
treat × week since adoption = 1	0.102** (0.042)	-0.014 (0.040)	-0.031 (0.085)
treat × week since adoption = 2	0.095** (0.043)	0.029 (0.041)	-0.161* (0.084)
treat × week since adoption = 3	0.087** (0.038)	0.013 (0.046)	-0.244*** (0.078)
treat × week since adoption = 4	0.147*** (0.046)	0.056 (0.043)	-0.290*** (0.107)
treat × week since adoption = 5	0.162*** (0.050)	0.050 (0.046)	-0.201** (0.098)
treat × week since adoption = 6	0.173*** (0.058)	0.038 (0.054)	-0.371*** (0.104)
treat × week since adoption = 7	0.213*** (0.049)	0.072 (0.052)	-0.396*** (0.117)
treat × week since adoption = 8	0.179*** (0.050)	0.026 (0.059)	-0.368*** (0.118)
treat × week since adoption = 9	0.222*** (0.061)	0.032 (0.067)	-0.276*** (0.092)
treat × week since adoption = 10	0.222*** (0.056)	0.095 (0.063)	-0.197* (0.117)
treat × week since adoption = 11	0.234*** (0.050)	0.096 (0.067)	-0.373*** (0.122)
treat × week since adoption = 12	0.206*** (0.051)	0.081 (0.068)	-0.246* (0.127)
treat × week since adoption = 13	0.218*** (0.050)	0.070 (0.079)	-0.328** (0.166)
treat × week since adoption = 14	0.263*** (0.063)	0.161** (0.066)	-0.238** (0.101)
<i>Fixed-effects</i>			
individual × subexperiment	✓	✓	✓
week since adoption	✓	✓	✓
<i>Fit statistics</i>			
Pseudo R ²	1.0932	1.2103	0.04440
Observations	30,048	30,048	29,994

Clustered (individual × subexperiment) standard-errors in parentheses
Signif. Codes: ***, 0.01, **, 0.05, *, 0.1

Table S12: Effect estimates of short-form video platforms on mobile usage metrics with a period length of 90 days prior to and post the adoption window.

Supplement C Robustness checks

C.1 Assessment of the parallel trends assumption

A central assumption for identification of the difference-in-differences (DiD) framework is that, absent treatment, treated and control individuals would have followed parallel trends in the outcome variable. Directly testing this assumption is not possible, since the counterfactual path of treated units is not observable. However, a feature of DiD frameworks with a leads-and-lags specification is that they allow researchers to examine pre-adoption effect estimates as a way of testing the plausibility of parallel trends. Specifically, if the parallel trends assumption is valid, there should not be systematic differences in outcome trends between treated and control units before adoption. By estimating treatment effects for these pre-adoption periods, we obtain a diagnostic check of whether treated and control groups were evolving similarly prior to adoption. Although these checks do not amount to a formal test of the parallel trends assumption, they represent a diagnostic available within the DiD framework. In our analyses, the pre-adoption tests are consistently fulfilled (see Figure 3). Across all outcomes considered, the estimated pre-adoption effects are close to zero in magnitude and statistically insignificant. Taken together, these results strengthen the robustness of our empirical model by providing evidence that the identifying parallel trends assumption is plausible in our application.

C.2 Effect of exposure to short-form videos

In our main analysis, we carefully constructed treatment and control groups to estimate the effect of short-form video platforms. We therefore defined adopters of TikTok as the treatment group and adopters of Instagram or Facebook as our control group, during a period before the latter introduced their own short-form video format. We thus ensured that both groups adopt comparably large platforms with similar dynamics, but differ mainly in their exposure to short-form videos. Here, we performed several robustness checks suggesting that the estimated effects in our main analyses mainly capture the impact of exposure to short-form videos rather than the general effect of adopting a social media application. Specifically, we defined two different control groups: (1) non-adopters of TikTok and (2) adopters of Instagram or Facebook during a period *after* the latter introduced their own short-form videos. Here, we assumed the following: (1) Compared to non-adopters, TikTok adopters had significantly higher durations, more sessions, and a lower average TAP post-adoption, as adopters were faced with the effect of a new social media application *and* exposure to short-form videos; (2) Compared to Instagram or Meta adopters (with short-form videos), TikTok adopters were not having significantly different durations, sessions, or average TAP, because both treatment and control groups were exposed to short-form videos.

The results highlight the robustness of our identification strategy (see Supplementary Figure S6). Specifically, when comparing TikTok adopters to non-adopters, we found significantly higher post-adoption mobile usage durations, an increase in the number of sessions, and a substantial decrease in average TAP. When comparing TikTok adopters to adopters of Instagram or Facebook during a period in which those platforms already featured short-form videos, we observed no consistent or significant differences across any of the three outcome metrics. This null finding supports our interpretation that the effects are mainly driven by exposure to short-form videos rather than by social media adoption more generally.

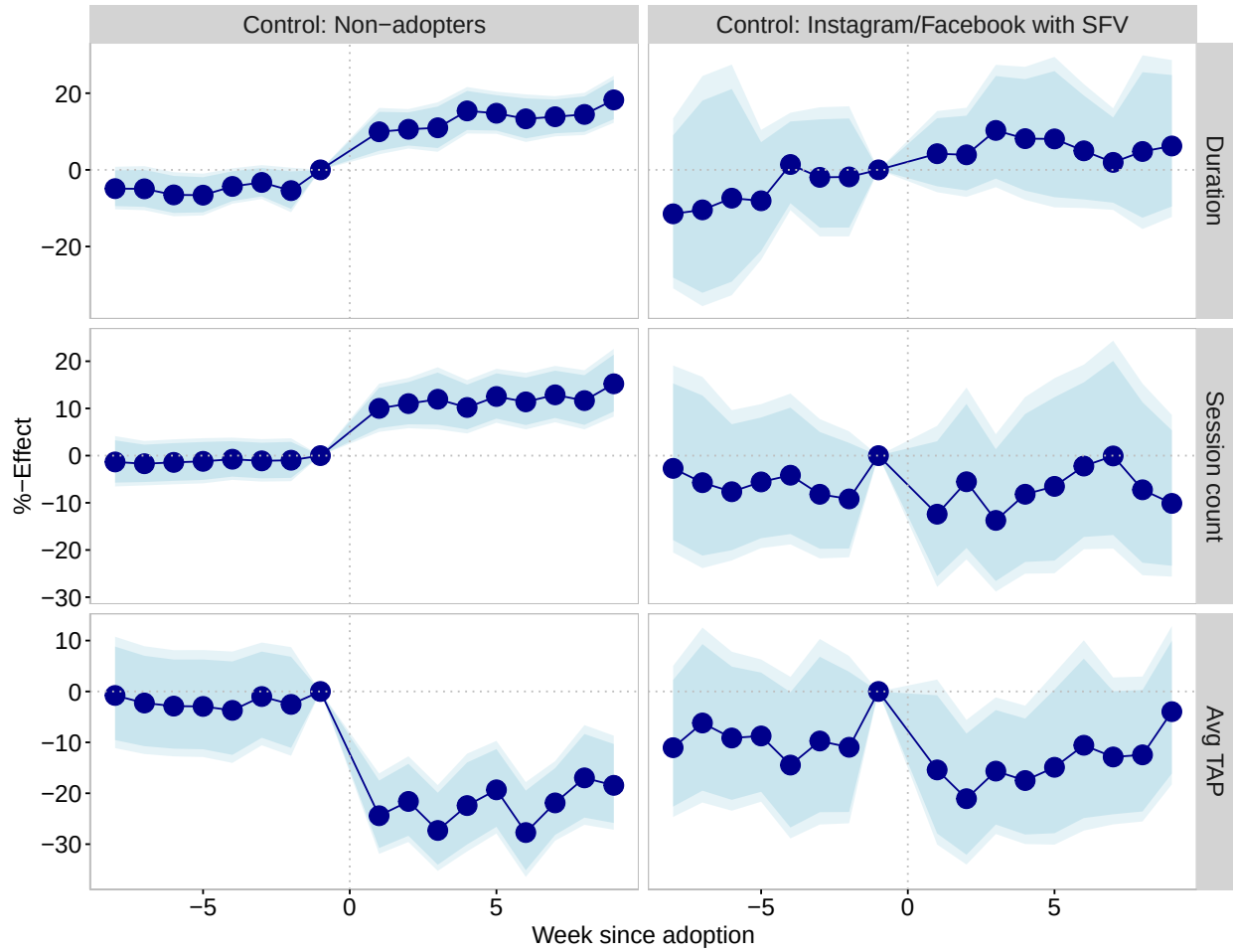


Figure S6: **Robustness check for the selected treatment and control group to estimate the effect of exposure to short-form videos.** We plot the aggregated effect estimates (in %) of exposure to short-form videos with two different control groups: (1) Non-adopters of TikTok and (2) Adopters of Instagram or Facebook during a period *after* the latter introduced their own short-form videos. The results suggest that our identification strategy isolates the effect of short-form videos, as the effects are stronger when comparing to non-adopters and vanish when comparing to Instagram or Facebook adopters during a period *after* the latter introduced their own short-form videos (SFV).

743 C.3 Alternative model specifications

744 We evaluated the robustness of our findings using alternative model specifications commonly ap-
 745 plied in the literature (e.g., [40,41,42]). These models differ in how they handle staggered treatment
 746 adoption, and therefore provide useful benchmarks to assess the sensitivity of our results to alter-
 747 native estimation strategies. First, we estimated a standard two-way fixed effects (TWFE) model
 748 with a leads-and-lags specification, which is the canonical dynamic DiD approach in applied work
 749 but has been shown to yield biased estimates under staggered adoption [43]. Second, we used the
 750 interaction-weighted (IW) estimator proposed in [43], which estimates conditional average treat-
 751 ment effects of the treated (CATTs) per adoption cohort and then aggregates them into dynamic
 752 treatment effects using weighted averages.

753 C.3.1 Two-way fixed-effects

754 As a first alternative, we estimated dynamic treatment effects using the standard two-way fixed
 755 effects (TWFE) framework with a leads-and-lags specification. Recent literature has shown that
 756 it may produce biased estimates in staggered adoption settings when individuals have different
 757 adoption timings [43]. Nevertheless, it remains a useful benchmark and allows us to assess the
 758 robustness of our results.

759 Formally, we estimated the following model:

$$\log(Y_{i,t}) = \sum_{j \in L} \beta_j D_{i,t}^j + \sum_{l \in L} \delta_l (T_i \times D_{i,t}^l) + \alpha_i + \lambda_t + \varepsilon_{i,t}, \quad (\text{S1})$$

760 where Y_{it} is the mobile usage outcome for individual i at week t , L denotes the set of relative
 761 weeks included in the specification,⁴ β_j captures the relative week fixed effects, $D_{i,t}^j$ is an indicator
 762 for individual i being j weeks away from adoption, T_i denotes treatment status, δ_l are the dynamic

⁴Similar to our stacked, dynamic specification in Equation (2), L includes the eight weeks before and nine weeks after adoption, except the omitted periods -1 and 0 to avoid multicollinearity.

treatment effects, α_i are individual fixed effects, and λ_t are calendar-week fixed effects. As before, the pre-adoption coefficients $\{\delta_l\}_{l<1}$ allow us to test the parallel trends assumption for pre-adoption periods, while the post-adoption coefficients $\{\delta_l\}_{l\geq 1}$ capture the dynamic effect of exposure to short-form videos. We use the same estimators for each mobile usage metric as described in Section 4. Hence, all estimated coefficients can be interpreted as percentage changes of the form $(\exp(\delta_l) - 1) \times 100 \%$.

The results highlight that our main findings are robust to the standard two-way fixed effects specification (see Table S13). The effect estimates are again significant and positive post-adoption for the outcome duration. Session counts also increased, with effects becoming significant from week 4 onward. At the same time, average TAP decreased, with particularly larger reductions at later weeks post-adoption. Pre-adoption coefficients remained close to zero and not significant across all three outcomes, supporting the validity of the parallel trends assumption for pre-adoption periods. Overall, these results closely mirror those obtained with our stacked, dynamic DiD framework, suggesting that the estimated effects are not sensitive to the choice of specification.

Dependent Variables: Distributions:	log(duration) Poisson	log(session count) Poisson	log(avg TAP) Gaussian
<i>Variables</i>			
treat × week since adoption = -8	-0.004 (0.038)	0.031 (0.039)	0.016 (0.064)
treat × week since adoption = -7	-0.010 (0.037)	0.010 (0.038)	-0.007 (0.061)
treat × week since adoption = -6	0.017 (0.035)	0.023 (0.033)	-0.040 (0.062)
treat × week since adoption = -5	-0.042 (0.037)	0.007 (0.032)	0.004 (0.060)
treat × week since adoption = -4	-0.027 (0.037)	0.010 (0.034)	-0.031 (0.065)
treat × week since adoption = -3	-0.040 (0.030)	0.012 (0.029)	0.004 (0.055)
treat × week since adoption = -2	-0.010 (0.030)	0.011 (0.026)	0.032 (0.056)
treat × week since adoption = 1	0.067* (0.036)	0.015 (0.030)	0.004 (0.058)
treat × week since adoption = 2	0.076** (0.032)	0.038 (0.034)	-0.009 (0.059)
treat × week since adoption = 3	0.063* (0.033)	0.047 (0.034)	-0.106 (0.065)
treat × week since adoption = 4	0.077** (0.038)	0.085** (0.038)	-0.111* (0.066)
treat × week since adoption = 5	0.111*** (0.042)	0.114*** (0.036)	-0.076 (0.075)
treat × week since adoption = 6	0.120** (0.048)	0.104** (0.045)	-0.170** (0.074)
treat × week since adoption = 7	0.136*** (0.038)	0.142*** (0.041)	-0.103 (0.069)
treat × week since adoption = 8	0.128*** (0.045)	0.091** (0.040)	-0.106 (0.079)
treat × week since adoption = 9	0.134*** (0.043)	0.118*** (0.041)	-0.245** (0.117)
<i>Fixed-effects</i>			
individual	✓	✓	✓
week	✓	✓	✓
week since adoption	✓	✓	✓
<i>Fit statistics</i>			
Pseudo R ²	1.0622	1.1334	0.05030
Observations	29,637	29,637	29,586

Clustered (individual) standard-errors in parentheses

Signif. Codes: ***, 0.01, **, 0.05, *, 0.1

Table S13: Effect estimates of short-form video platforms on mobile usage metrics using the standard two-way fixed effects DiD with a leads-and-lags specification.

C.3.2 Interaction-weighted estimator

As a second alternative, we applied the interaction-weighted (IW) estimator proposed in [43], which, similarly to our stacked approach, addresses the limitations of standard two-way fixed effects models in staggered adoption settings. The IW estimator avoids the problematic comparison across individuals with different adoption timings by first estimating conditional average treatment effects of the treated (CATTs) for each cohort. Here, individuals were categorized into different cohorts based on their initial treatment timing (i.e., the adoption week). The CATTs were then aggregated into dynamic treatment effects using weighted averages.

Formally, the procedure consisted of three steps:

Step 1. We first estimated cohort-specific CATTs by interacting relative week indicators with cohort indicators in a two-way fixed effects regression of the form

$$\log(Y_{i,t}) = \sum_{j \in L} \beta_j D_{i,t}^j + \sum_{e \in E} \sum_{l \in L} \delta_{e,l} \cdot (T_i \times G_i^e \times D_{i,t}^l) + \alpha_i + \lambda_t + \epsilon_{i,t}, \quad (\text{S2})$$

where Y_{it} is the mobile usage outcome for individual i in calendar week t , L denotes the set of relative weeks included in our specification,⁵ $D_{i,t}^j$ is an indicator for individual i being j weeks away from adoption, E denotes the set of cohorts, T_i denotes treatment status, G_i^e is an indicator for individual i being in cohort e , $\delta_{e,l}$ capture the effect for cohort e at relative time l , i.e., $\text{CATT}_{e,l}$, α_i are individual fixed effects, and λ_t are calendar-week fixed effects. We use the same estimators for each mobile usage metric as described in Section 4. Hence, all estimated coefficients can be interpreted as percentage changes of the form $(\exp(\delta_{e,l}) - 1) \times 100 \%$.

Step 2. We then computed weights $w(e, l)$ as the share of individuals at relative time l that belong to cohort e .

Step 3. Lastly, we aggregated the cohort-specific dynamic effect estimates to overall dynamic

⁵Similar to our stacked, dynamic specification in Equation (2), L includes the eight weeks before and nine weeks after adoption, except the omitted periods -1 and 0 to avoid multicollinearity.

798 treatment effects using the weighted average

$$\hat{\delta}_l = \frac{1}{|L|} \sum_{l \in L} \sum_{e \in E} \hat{\delta}_{e,l} \hat{w}(e, l), \quad (\text{S3})$$

799 where $\hat{\delta}_{e,l}$ is the cohort-specific dynamic effect estimate from step 1 and $\hat{w}(e, l)$ is the estimated
800 weight from step 2.

801 The results based on the IW estimator from [43] (see Supplementary Table S14) closely align
802 with those from our specification in Equation (2). We again observed increases in mobile usage du-
803 ration and session counts, and reductions in average TAP post-adoption. Importantly, pre-adoption
804 coefficients remained close to zero, consistent with the parallel trends assumption for pre-adoption
805 periods. These findings confirm that our main results are not driven by the choice of estimator and
806 remain robust when using the IW estimator.

Dependent Variables: Distributions:	log(duration) Poisson	log(session count) Poisson	log(avg TAP) Gaussian
<i>Variables</i>			
treat × week since adoption = -8	0.004 (0.039)	0.024 (0.040)	0.030 (0.054)
treat × week since adoption = -7	0.001 (0.038)	0.012 (0.034)	-0.006 (0.055)
treat × week since adoption = -6	0.020 (0.035)	0.018 (0.031)	-0.012 (0.052)
treat × week since adoption = -5	-0.039 (0.039)	0.002 (0.028)	0.017 (0.052)
treat × week since adoption = -4	-0.027 (0.038)	0.003 (0.028)	-0.037 (0.058)
treat × week since adoption = -3	-0.037 (0.031)	0.0010 (0.026)	0.034 (0.044)
treat × week since adoption = -2	-0.008 (0.029)	0.004 (0.023)	0.056 (0.044)
treat × week since adoption = 1	0.066* (0.038)	-0.002 (0.028)	0.031 (0.047)
treat × week since adoption = 2	0.077** (0.034)	0.026 (0.032)	0.030 (0.048)
treat × week since adoption = 3	0.071** (0.033)	0.038 (0.029)	-0.054 (0.052)
treat × week since adoption = 4	0.080** (0.038)	0.072* (0.037)	-0.099* (0.053)
treat × week since adoption = 5	0.110*** (0.042)	0.098*** (0.037)	-0.035 (0.068)
treat × week since adoption = 6	0.126*** (0.047)	0.092** (0.045)	-0.113* (0.060)
treat × week since adoption = 7	0.149*** (0.038)	0.133*** (0.043)	-0.049 (0.079)
treat × week since adoption = 8	0.138*** (0.045)	0.079* (0.041)	-0.067 (0.071)
treat × week since adoption = 9	0.151*** (0.043)	0.111*** (0.043)	-0.230*** (0.082)
<i>Fixed-effects</i>			
individual	✓	✓	✓
week	✓	✓	✓
week since adoption	✓	✓	✓
<i>Fit statistics</i>			
Pseudo R ²	1.0604	1.1295	0.05731
Observations	29,637	29,637	29,586

Clustered (individual) standard-errors in parentheses
Signif. Codes: ***, 0.01, **, 0.05, *, 0.1

Table S14: Effect estimates of short-form video platforms on mobile usage metrics using the interaction-weighted estimator [43].

C.4 Sensitivity analyses

C.4.1 Causal sensitivity analysis

To further assess the robustness of our estimates, we conducted a causal sensitivity analysis following [68]. This analysis complements our prior robustness checks by addressing the possibility of *time-varying unobserved confounding* that could bias our effects in the post-adoption periods.

Specifically, the validity of the DiD framework relies on the parallel trends assumption and hence allows for time-invariant unobserved confounding. Here, the parallel trends assumption is assessable via pre-adoption coefficient estimates as evaluated in Supplement C.1. Hence, this causal sensitivity analysis evaluates whether unobserved confounders varying over time could affect our results.

We focused on the two mobile usage metrics that showed significant effects in the main analyses, namely, mobile usage duration and average TAP. For each metric, we first compute the partial R^2 of the treatment with the metric, denoted $R^2 Y \sim D_{post} | X$. We then computed two so-called robustness values (see [68]): (1) $RV_{q=1}$ indicates the minimum strength of association that a time-varying unobserved confounder would need to have with both the treatment and the metric to fully explain away the estimated post-adoption effect; (2) $RV_{q=1, \alpha=0.05}$ indicates the strength required to render the effect statistically insignificant at the 5% level.

The results are summarized in Supplementary Table S15. For both outcomes, the robustness values indicated that a time-varying unobserved confounder would need to have a substantially stronger association with both the treatment and the outcome than the observed treatment itself to alter our conclusions. Specifically, for mobile usage duration, $RV_{q=1}$ and $RV_{q=1, \alpha=0.05}$ are 11.08% and 10.06%, respectively; for average TAP, the corresponding values are 13.98% and 12.99%. Compared to the partial $R^2_{Y \sim D | X}$, this implied that a confounder would need to be substantially more associated to both the treatment and the outcome to explain away or render insignificant our estimated effects. Such levels of confounding are substantially high in our setting, suggesting that the post-adoption effects we estimate are robust to potential time-varying unobserved confounding.

Outcome	$R^2_{Y \sim D X}$	$RV_{q=1}$	$RV_{q=1, \alpha=0.05}$
Duration	1.36%	11.08%	10.06%
Avg TAP	2.22%	13.98%	12.99%

$R^2_{Y \sim D_{post}|X}$ is the partial R^2 of the treatment with the outcome

Table S15: **Effect estimates of short-form video platforms on mobile usage metrics using the interaction-weighted estimator [43].**

C.4.2 Placebo test

We performed placebo tests to assess potential anticipation effects prior to adoption. Specifically, the DiD framework relies on the *no-anticipation* assumption, which requires that the treatment has no effect before its official implementation. To test this, we conducted a placebo test by assigning a *fake* adoption date that precedes the actual adoption. If users systematically altered their mobile usage before the real adoption, e.g., in anticipation of using short-form video platforms, this test would yield a significant placebo effect. We therefore re-estimated both the propensity score weighting models using pre-adoption mobile activity corresponding to the new placebo periods and our stacked, dynamic DiD model using placebo periods.

when we shifted the placebo adoption week to -1 , -2 , -3 , and -4 relative to the true adoption week, the placebo estimates were small in magnitude and generally statistically indistinguishable from zero across all three outcome variables (see Supplementary Tables S16 to S19). This indicates that the observed post-adoption effects are unlikely to be driven by anticipatory behavior and instead emerge following the actual adoption of short-form video platforms.

Dependent Variables: Distributions:	log(duration) Poisson	log(session count) Poisson	log(avg TAP) Gaussian
<i>Variables</i>			
treat × week since adoption = -8	-0.011 (0.048)	-0.035 (0.052)	0.079 (0.078)
treat × week since adoption = -7	-0.068 (0.048)	-0.049 (0.046)	-0.004 (0.065)
treat × week since adoption = -6	-0.023 (0.039)	-0.025 (0.041)	0.037 (0.066)
treat × week since adoption = -5	-0.020 (0.038)	-0.034 (0.042)	-0.003 (0.069)
treat × week since adoption = -4	0.002 (0.036)	-0.015 (0.042)	-0.074 (0.075)
treat × week since adoption = -3	-0.016 (0.032)	-0.012 (0.043)	0.001 (0.065)
treat × week since adoption = -1	0.009 (0.028)	0.007 (0.039)	-0.110* (0.061)
treat × week since adoption = 1	0.089** (0.038)	-0.007 (0.039)	-0.022 (0.070)
treat × week since adoption = 2	0.108*** (0.041)	0.010 (0.044)	-0.071 (0.068)
treat × week since adoption = 3	0.095** (0.040)	0.026 (0.043)	-0.198*** (0.076)
treat × week since adoption = 4	0.128*** (0.043)	0.082* (0.048)	-0.141** (0.072)
treat × week since adoption = 5	0.181*** (0.043)	0.088* (0.051)	-0.085 (0.075)
treat × week since adoption = 6	0.145*** (0.050)	0.042 (0.053)	-0.155** (0.079)
treat × week since adoption = 7	0.167*** (0.038)	0.073 (0.051)	-0.178** (0.085)
treat × week since adoption = 8	0.178*** (0.048)	0.051 (0.054)	-0.096 (0.077)
treat × week since adoption = 9	0.160*** (0.049)	0.076 (0.052)	-0.195** (0.081)
<i>Fixed-effects</i>			
individual × subexperiment	✓	✓	✓
week since adoption	✓	✓	✓
<i>Fit statistics</i>			
Pseudo R ²	1.0635	1.1357	0.05018
Observations	29,637	29,637	29,586

Clustered (individual × subexperiment) standard-errors in parentheses

Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

Table S16: Effect estimates from placebo difference-in-differences analyses, shifting the treatment week to -1 relative to the actual adoption week. Placebo periods are highlighted in bold.

Dependent Variables: Distributions:	log(duration) Poisson	log(session count) Poisson	log(avg TAP) Gaussian
<i>Variables</i>			
treat × week since adoption = -8	-0.001 (0.044)	-0.042 (0.048)	0.033 (0.065)
treat × week since adoption = -7	-0.041 (0.048)	-0.049 (0.046)	0.003 (0.064)
treat × week since adoption = -6	0.005 (0.046)	-0.027 (0.040)	-0.004 (0.065)
treat × week since adoption = -5	-0.037 (0.041)	-0.046 (0.036)	-0.018 (0.062)
treat × week since adoption = -4	-0.002 (0.039)	-0.032 (0.034)	-0.076 (0.070)
treat × week since adoption = -2	-0.010 (0.034)	-0.065** (0.033)	-0.024 (0.060)
treat × week since adoption = -1	0.030 (0.035)	-0.018 (0.034)	-0.085 (0.056)
treat × week since adoption = 1	0.066** (0.034)	-0.047 (0.037)	-0.045 (0.061)
treat × week since adoption = 2	0.086** (0.040)	-0.028 (0.043)	-0.059 (0.063)
treat × week since adoption = 3	0.067* (0.037)	-0.014 (0.041)	-0.200*** (0.068)
treat × week since adoption = 4	0.084** (0.042)	0.040 (0.054)	-0.210*** (0.079)
treat × week since adoption = 5	0.092** (0.041)	0.059 (0.054)	-0.156 (0.101)
treat × week since adoption = 6	0.082* (0.046)	0.063 (0.057)	-0.214*** (0.071)
treat × week since adoption = 7	0.145*** (0.038)	0.086 (0.058)	-0.229*** (0.084)
treat × week since adoption = 8	0.124*** (0.044)	0.058 (0.058)	-0.194* (0.103)
treat × week since adoption = 9	0.135*** (0.046)	0.061 (0.057)	-0.213*** (0.080)
<i>Fixed-effects</i>			
individual × subexperiment	✓	✓	✓
week since adoption	✓	✓	✓
<i>Fit statistics</i>			
Pseudo R ²	1.0677	1.1285	0.04911
Observations	29,637	29,637	29,586

Clustered (individual × subexperiment) standard-errors in parentheses

Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

Table S17: Effect estimates from placebo difference-in-differences analyses, shifting the treatment week to −2 relative to the actual adoption week. Placebo periods are highlighted in bold.

Dependent Variables: Distributions:	log(duration) Poisson	log(session count) Poisson	log(avg TAP) Gaussian
<i>Variables</i>			
treat × week since adoption = -8	-0.009 (0.041)	-0.092* (0.053)	0.014 (0.066)
treat × week since adoption = -7	-0.021 (0.042)	-0.084** (0.038)	0.011 (0.058)
treat × week since adoption = -6	0.041 (0.038)	-0.052 (0.039)	-0.023 (0.055)
treat × week since adoption = -5	-0.014 (0.034)	-0.048 (0.036)	-0.026 (0.058)
treat × week since adoption = -3	0.003 (0.032)	-0.009 (0.039)	-0.096 (0.061)
treat × week since adoption = -2	0.010 (0.038)	-0.024 (0.036)	-0.065 (0.062)
treat × week since adoption = -1	0.011 (0.034)	-0.016 (0.038)	-0.101 (0.064)
treat × week since adoption = 1	0.072** (0.037)	-0.042 (0.042)	-0.062 (0.069)
treat × week since adoption = 2	0.080** (0.039)	-0.025 (0.044)	-0.075 (0.066)
treat × week since adoption = 3	0.073* (0.040)	0.013 (0.040)	-0.181** (0.072)
treat × week since adoption = 4	0.097** (0.039)	0.059 (0.055)	-0.187** (0.086)
treat × week since adoption = 5	0.129*** (0.046)	0.091 (0.056)	-0.155* (0.082)
treat × week since adoption = 6	0.130** (0.050)	0.045 (0.058)	-0.240*** (0.084)
treat × week since adoption = 7	0.202*** (0.045)	0.093 (0.058)	-0.221** (0.092)
treat × week since adoption = 8	0.175*** (0.048)	0.076 (0.058)	-0.148* (0.079)
treat × week since adoption = 9	0.191*** (0.044)	0.055 (0.056)	-0.178** (0.078)
<i>Fixed-effects</i>			
individual × subexperiment	✓	✓	✓
week since adoption	✓	✓	✓
<i>Fit statistics</i>			
Pseudo R ²	1.0670	1.1636	0.05097
Observations	29,637	29,637	29,586

Clustered (individual × subexperiment) standard-errors in parentheses

Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

Table S18: Effect estimates from placebo difference-in-differences analyses, shifting the treatment week to −3 relative to the actual adoption week. Placebo periods are highlighted in bold.

Dependent Variables: Distributions:	log(duration) Poisson	log(session count) Poisson	log(avg TAP) Gaussian
<i>Variables</i>			
treat × week since adoption = -8	0.003 (0.046)	-0.037 (0.055)	0.032 (0.066)
treat × week since adoption = -7	0.027 (0.051)	-0.027 (0.052)	0.039 (0.062)
treat × week since adoption = -6	0.024 (0.043)	-0.007 (0.040)	0.016 (0.065)
treat × week since adoption = -4	0.026 (0.039)	0.007 (0.043)	-0.071 (0.075)
treat × week since adoption = -3	0.026 (0.038)	-0.022 (0.051)	-0.129* (0.072)
treat × week since adoption = -2	0.069 (0.043)	-0.028 (0.043)	-0.071 (0.075)
treat × week since adoption = -1	0.031 (0.043)	0.025 (0.045)	-0.053 (0.068)
treat × week since adoption = 1	0.073 (0.046)	-0.005 (0.045)	-0.104 (0.086)
treat × week since adoption = 2	0.074 (0.053)	0.023 (0.048)	-0.086 (0.085)
treat × week since adoption = 3	0.095 (0.059)	0.026 (0.050)	-0.272*** (0.096)
treat × week since adoption = 4	0.122** (0.051)	0.074 (0.050)	-0.190** (0.083)
treat × week since adoption = 5	0.166*** (0.053)	0.097* (0.051)	-0.139* (0.080)
treat × week since adoption = 6	0.163*** (0.057)	0.060 (0.055)	-0.212** (0.082)
treat × week since adoption = 7	0.224*** (0.057)	0.086 (0.053)	-0.271*** (0.097)
treat × week since adoption = 8	0.207*** (0.054)	0.043 (0.057)	-0.150 (0.094)
treat × week since adoption = 9	0.204*** (0.065)	0.058 (0.052)	-0.222** (0.095)
<i>Fixed-effects</i>			
individual × subexperiment	✓	✓	✓
week since adoption	✓	✓	✓
<i>Fit statistics</i>			
Pseudo R ²	1.0764	1.1506	0.05013
Observations	29,637	29,637	29,586

Clustered (individual × subexperiment) standard-errors in parentheses
Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

Table S19: Effect estimates from placebo difference-in-differences analyses, shifting the treatment week to -4 relative to the actual adoption week. Placebo periods are highlighted in bold.